

Yang-Mills Theory

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Chapter 1 Geometry and Topology of 4-manifolds

1 Gluing ASD Connections

In this section, we consider the following question: suppose D_1, D_2 are ASD connections on bundle E_1, E_2 over compact, oriented 4-manifolds M_1, M_2 respectively. It is well known that this two manifolds can be glued as $M_1 \# M_2$. If there exists some excesses to glue two bundles E_1, E_2 , denoted by $E_1 \# E_2$, can we construct a new ASD connection on ‘bundle’ $E_1 \# E_2$ over $M_1 \# M_2$?

Before we glue two ASD connections, we should define the bundle $E_1 \# E_2$ in a legitimate way. We first introduce the connected sum $M = M_1 \# M_2$. Choose points $x_1 \in M_1$ and $x_2 \in M_2$, suppose for simplicity that the metrics on M_1 and M_2 are flat in neighbourhoods of x_1 and x_2 respectively. We can identify neighbourhoods of x_1, x_2 in M_1, M_2 with neighbourhoods of zero in tangent space $T_{x_1}M_1, T_{x_2}M_2$ through these flat metrics. Now let

$$\sigma : T_{x_1}M_1 \rightarrow T_{x_2}M_2$$

Be an orientation-reversing linear isometry. For any $\lambda > 0$ we define

$$f_\lambda : T_{x_1}M_1 \setminus 0 \rightarrow T_{x_2}M_2 \setminus 0$$

$$\xi \mapsto \frac{\lambda}{|\xi|^2} \sigma(\xi)$$

Let $\Omega_i \subset M_i$ be the annulus centred on x_i with inner radius $N^{-1}\lambda^{1/2}$ and outer radius $N\lambda^{1/2}$ (N to be determined s.t. $N\lambda^{1/2} \ll 1$), then f_λ induces a diffeomorphism from Ω_1 to Ω_2 . Denote the open set obtained by removing the $N^{-1}\lambda^{1/2}$ ball about x_i by M'_i , then we can define the connected sum of manifolds as below:

$$M = M'_1 \cup_{f_\lambda} M'_2$$

Where the annulus Ω_i are identified by f_λ . Since f_λ is a conformal map, M has a natural conformal structure, depending upon the parameters σ, λ (but independent of N). More specifically, the attaching function f_λ we constructed must have the following two properties:

- be conformal, thereupon there exists conformal structure on base manifold M , then we can fix one so that there is no confuse on whether the connection is SD or ASD.
- be orientation-preserving, since a connection is ASD or SD depends on the orientation. Therefore the function σ must be orientation-reversing, since inversion map is orientation-reversing.

Now we turn to bundles E_i over M_i and connections D_i . We hope to replace D_i by new connections D'_i which are flat in neighbourhoods of x_i . Choose a G-isomorphism of the fibers:

$$\rho : (E_1)_{x_1} \rightarrow (E_2)_{x_2}$$

Using the flat connection D'_i we can spread this isomorphism out to give a bundle isomorphism g_ρ between E_i over the annulus Ω_i . Thus we can define a bundle $E(\rho)$ through the above identification. Moreover, there is an induced connection on $E(\rho)$, denoted by $D'(\rho)$. Unfortunately, $D'(\rho)$ is not a ASD connection, however very close to it. We will construct a nearby ASD connection. (in the residue of this section, we simplify $D'(\rho), E(\rho)$ as D', E respectively, since ρ fixed)

Back to definition of ASD connections. If we fix a reference connection D_0 , then any other connection D can be written as $D = D_0 + A$, where A is a section of $\Lambda^1(M) \otimes \text{ad}(E)$. Then curvature F_D of D can be computed as:

$$F_D = F_{D_0} + dA + A \wedge A$$

In the case D_0 is anti-self-dual, i.e. $(F_{D_0})^+ = 0$, we obtain

$$(F_D)^+ = (F_{D_0} + dA + A \wedge A)^+ = d^+A + (A \wedge A)^+$$

Hence D is anti-self-dual iff

$$d^+A + (A \wedge A)^+ = 0$$

Moreover, d^+ is the linearization of $(F_D)^+$. Therefore, if we denote the ASD connection on $E(\rho)$ by $E(\rho) = D' + a$, then we need to determine $a \in \Omega^1(E(\rho))$ s.t. the following equation

$$d_{D'}^+a + (a \wedge a)^+ = -F_{D'}^+ \tag{1.1}$$

Holds. Outline of the above process can be summarized as:

Step1 prove the map $d_{D'}^+$ has a inverse, denoted by P ,

Step2 seek a solution in the form $a = P(\xi)$, hence equation (1.1) can be written as $\xi + (P\xi \wedge P\xi)^+ = -F_{D'}^+$,

Step3 for specific operator S , study equation of the form $\xi + S(\xi) = \eta$, then deduce conclusion.

Step1 $d_{D'}^+ : L^q(\Omega_1(E)) \rightarrow L^p(\Omega_2^+(E))$ **has a right inverse.**

We first consider the inverse of $d_{D_1}^+$ on M_1 . Let $d_{D_1}^*$ be the adjoint of $d_{D_1}^+$, set $\Delta = d_{D_1}^+ \circ d_{D_1}^* : W^{2,p}(\Omega_2^+(E_1)) \rightarrow L^p(\Omega_2^+(E_1))$. By definition, for $\alpha \in \Omega_1, \beta \in \Omega_2^+$,

$$\begin{aligned} (d_{D_1}^+ \alpha, \beta) &= (d_{D_1} \alpha, \beta) = (\alpha, \delta_{D_1} \beta) \\ \Rightarrow d_{D_1}^* &= \delta_{D_1} \end{aligned}$$

Then for $\omega \in \Omega_2^+$,

$$\begin{aligned} \Delta \omega &= d_{D_1}^+ \circ d_{D_1}^* \omega = d_{D_1}^+ \circ (-\star d_{D_1} \star) \omega = -(d_{D_1} \star d_{D_1} \omega)^+ \\ &= (\star \delta_{D_1} d_{D_1} \omega)^+ = (\delta_{D_1} d_{D_1} \omega)^+ = \frac{1}{2}(\Delta_{\text{Hodge}} \omega)^+ \end{aligned}$$

Here we use $\delta = -\star d\star$ and $\star^2 = 1$. We have deduced $\Delta = 1/2(\Delta_{\text{Hodge}})^+$, furthermore Δ_{Hodge} commutes with $\star$, therefore, Δ is the restriction of Δ_{Hodge} . Assume $H_{D_1}^2 = 0$, then $H_+^2 = 0$ and $\Delta : W^{2,p}(\Omega_2^+(E_1)) \rightarrow L^p(\Omega_2^+(E_1))$ is invertible (denote its inverse by Δ^{-1}), this implies $d_{D_1}^+$ is surjective, and the right inverse of $d_{D_1}^+$ can be written as

$$P_1 \stackrel{\text{set}}{=} d_{D_1}^* \cdot \Delta^{-1}$$

The facts we need are P_1 is a bounded operator between Sobolev spaces, combined with the Sobolev Embedding Theorem we get

$$\|P_1 \omega\|_{L^q(\Omega_1(E_1))} \leq C_1 \|\omega\|_{L^p(\Omega_2^+(E_1))} \quad (1.2)$$

Where $1/q = 1/p - 1/4$. We have similar consequence for $d_{D_2}^+$ and D_2, P_2 .

Till now we have found inverse of $d_{D_1}^+, d_{D_2}^+$, which can be treated as part of the whole goal, we still owe to dispose the situation in the place where gluing happens. For this motivation, we attempt to construct cut-off functions as below:

- take functions γ_1, γ_2 defined on $M = M_1 \sharp M_2$, which satisfied $\gamma_1 + \gamma_2 = 1$; γ_i supported in $M_i \setminus B_{\frac{1}{2}\lambda^{1/2}}(x_i)$ and equal to 1 on $M_i \setminus B_{2\lambda^{1/2}}(x_i)$.
- take function β_1 defined on M_1 satisfied equal to 1 on $M_1 \setminus B_{N\lambda^{1/2}}(x_1)$ and supported in $M_1 \setminus B_{N^{-1}\lambda^{1/2}}(x_1)$; take β_2 defined on M_2 likewise.

The essential property of β_i is that $\int |\nabla \beta_i|^4$ can be choosen as small as we need by choosing N large. This situation is possible.

Thanks to the above cut-off functions, we can define a gobal defined operator via P_1, P_2 , in this regard

$$P\omega \stackrel{\text{set}}{=} \beta_1 P_1 \gamma_1 \omega + \beta_2 P_2 \gamma_2 \omega$$

Is operator P a right inverse of $d_{D'}^+$? Unfortunately, the answer is wrong, however, very close. Now we establish estimate estimates on $\|P\omega\|_{L^q(\Omega_1(E))}$.

Given a section ω in $\Omega_2^+(E)$, write $\omega = \gamma_1\omega + \gamma_2\omega$. Notice that $\gamma_1\omega$ can be regarded as a section of $\Omega_2^+(E_1)$, then (1.2) implies

$$\|P_1\gamma_1\omega\|_{L^q(\Omega_1(E_1))} \leq C_1 \|\gamma_1\omega\|_{L^p(\Omega_2^+(E))} \quad (1.3)$$

Obviously

$$\|\beta_1 P_1 \gamma_1 \omega\|_{L^q(\Omega_1(E_1))} \leq \|P_1 \gamma_1 \omega\|_{L^q(\Omega_1(E_1))} \quad (1.4)$$

From our setting, inequality $g' \geq g_1$ holds on support of β_1 , which implies

$$\|\beta_1 P_1 \gamma_1 \omega\|_{L^q(\Omega_1(E))} \leq \|\beta_1 P_1 \gamma_1 \omega\|_{L^q(\Omega_1(E_1))} \quad (1.5)$$

Combine (1.3), (1.4) and (1.5) we get

$$\|P\omega\|_{L^q(\Omega_1(E))} \leq (C_1 + C_2) \|\omega\|_{L^p(\Omega_2^+(E))} \quad (1.6)$$

Next we consider $d_{D'}^+ P\omega - \omega$ (recall that on $\text{supp}(\beta_i)$, $D' = D_i + a_i$):

$$\begin{aligned} \|d_{D'}^+ P\omega - \omega\|_{L^p(\Omega_2^+(E))} &\leq \|d_{D'}^+ \beta_1 P_1 \gamma_1 \omega - \gamma_1 \omega\|_{L^p(\Omega_2^+(E))} + \|d_{D'}^+ \beta_2 P_2 \gamma_2 \omega - \gamma_2 \omega\|_{L^p(\Omega_2^+(E))} \\ &= \|a_1 \sharp \beta_1 P_1 \gamma_1 \omega + d_{D_1}^+ \beta_1 P_1 \gamma_1 \omega - \gamma_1 \omega\|_{L^p} + (\dots) \\ &= \|a_1 \sharp \beta_1 P_1 \gamma_1 \omega + \nabla \beta_1 \sharp P_1 \gamma_1 \omega + \beta_1 d_{D_1}^+ P_1 \gamma_1 \omega - \gamma_1 \omega\|_{L^p} + (\dots) \\ &= \|a_1 \sharp \beta_1 P_1 \gamma_1 \omega + \nabla \beta_1 \sharp P_1 \gamma_1 \omega\|_{L^p} + (\dots) \\ &\leq \|a_1\|_{L^4} \|\beta_1 P_1 \gamma_1 \omega\|_{L^q} + \|\nabla \beta_1\|_{L^4} \|P_1 \gamma_1 \omega\|_{L^q} + (\dots) \\ &\leq \varepsilon \|\omega\|_{L^p(\Omega_2^+(E))} \end{aligned}$$

The coefficient ε can be arbitrarily small so long as we choose b small enough and N large enough. Based on the above, we write

$$d_{D'}^+ P = I + R$$

Where the operator norm of R small. Therefore $I + R$ has a inverse, denote by Q , s.t.

$$d_{D'}^+ (PQ) = I$$

Hence $PQ : L^p(\Omega_2^+(E)) \rightarrow L^q(\Omega_1(E))$ is the right inverse of $d_{D'}^+$, we need.

Step2&Step3 Modify equation (1.1) and solve it by Contraction Mapping Principle

We have instructed that for $D = D' + a$, we need to determine $a \in \Omega^1(E)$ such that the following equation holds:

$$d_{D'}^+ a + (a \wedge a)^+ = -F_{D'}^+$$

holds. Still denote the right inverse (PQ) above by P , it suffice to find $\xi \in L^p(\Omega_2^+(E))$, s.t. $a = P\xi$ together with

$$\begin{aligned} d_{D'}^+ P\xi + (P\xi \wedge P\xi)^+ &= -F_{D'}^+ \\ \text{i.e. } \xi + (P\xi \wedge P\xi)^+ &= -F_{D'}^+. \end{aligned} \tag{1.7}$$

We introduce a lemma:

Lemma 1.1. *Let $S : B \rightarrow B$ be a smooth map on a Banach space with $S(0) = 0$ and for some $C > 0$,*

$$\|S\xi_1 - S\xi_2\| \leq C(\|\xi_1\| + \|\xi_2\|)(\|\xi_1\| - \|\xi_2\|)$$

holds for all $\xi_1, \xi_2 \in B$. Then $\exists \delta > 0$, s.t. for each $\eta \in B$ with $\|\eta\| < \delta$, there is a ξ with $\|\xi\| < \delta$ s.t.

$$\xi + S(\xi) = \eta$$

Proof. This lemma is proved by Contraction Mapping Principle. Define a new operator

$$\begin{aligned} T : B &\rightarrow B \\ \xi &\mapsto \eta - S\xi \end{aligned}$$

Then equation becomes $T\xi = \xi$, i.e. the fix point equaiton of map T . Since

$$\|T\xi_1 - T\xi_2\| = \|S\xi_1 - S\xi_2\| \leq C(\|\xi_1\| + \|\xi_2\|)(\|\xi_1\| - \|\xi_2\|)$$

Take δ_1 small s.t. if $\|\xi_1\|, \|\xi_2\| < \delta_1$, T is contractive. Since

$$\|T\xi\| \leq \|\eta\| + \|S\xi\| \leq \|\eta\| + C\|\xi\|^2$$

Choose $\delta_2 \leq \min\{1/2C, \delta_1\}$, then for $\|\xi\| \leq \delta_2, \eta \leq \delta_2/2$ we have

$$\|T\xi\| \leq \frac{\delta_2}{2} + C\delta_2^2 \leq \frac{\delta_2}{2} + C\frac{1}{2C}\delta_2 = \delta_2$$

Therefore, T maps $\{\|\xi\| \leq \delta_2\}$ to itself, and it is contractive meanwhile, then Contraction Mapping Principle implies this lemma. □

Back to Step2&Step3, set $S(\xi) = (P\xi \wedge P\xi)^+$, then

$$\begin{aligned} \|S\xi\|_{L^p(\Omega_2^+(E))} &\leq C\|P\xi\|_{L^q(\Omega_1(E))}\|P\xi\|_{L^4(\Omega_1(E))} \\ S\xi_1 - S\xi_2 &= (P\xi_1 \wedge P\xi_1 - P\xi_2 \wedge P\xi_2) \\ &= (P\xi_1 \wedge (P\xi_1 - P\xi_2) + (P\xi_1 - P\xi_2) \wedge P\xi_2)^+ \\ &= ((P\xi_1 + P\xi_2) \wedge (P\xi_1 - P\xi_2))^+ \end{aligned}$$

by boundness of P and Holder's inequality, there exists a constant C , s.t.

$$\|S\xi_1 - S\xi_2\| \leq C(\|\xi_1\| + \|\xi_2\|)(\|\xi_1\| - \|\xi_2\|)$$

Holds for all $\xi_1, \xi_2 \in L^p(\Omega_2^+(E))$. So far, to apply above lemma, it remains to check $\|F_{D'}^+\|_{L^p(\Omega_2^+(E))}$ can be arbitrarily small. This is obvious since $\|F_{D'}^+\|_L^\infty \leq \text{const}$ and supported in $B_b(x_1) \setminus B_{b/2}(x_1)$ (in M_1) and $B_b(x_2) \setminus B_{b/2}(x_2)$ (in M_2), choose b small is all we need.

Eventually, we find a solution of equation (1.1), i.e. an ASD connection on bundle $E = E_1 \# E_2$ over $M = M_1 \# M_2$.

To sum up, we obtain the following theorem:

Theorem 1.2. *Suppose D_1, D_2 are ASD connections on E_1, E_2 over M_1, M_2 with $H_{D_1}^2 = H_{D_2}^2 = 0$, then there is an $W^{1,2}$ ASD connection on bundle $E = E_1 \# E_2$ over $M = M_1 \# M_2$.*

2 Integrability Theorem for Connections

Theorem 1.3. *If E is a bundle over H and A is a flat connection on E there is a bundle isomorphism taking E to the trivial bundle over H and A to the product connection.*

3 Application of Gauge Theory to 4-Dimensional Topology

This section mainly concerned on Donaldson's paper, "Application of Gauge Theory to Four Dimensional Topology", and we will state the proof of the following theorem:

Theorem 1.4. *Let X be a compact smooth simply connected oriented 4-manifold with the property that the associated form Q is positive definite. Then that form is equivalent, over the integers, to the standard diagonal form, so in some base:*

$$Q(u_1, u_2, \dots, u_r) = u_1^2 + u_2^2 + \dots + u_r^2.$$

Chapter 2 The Analytical Study of Yang-Mills Theory

1 Gauge Fixing Theorem

Theorem 2.1. *Let D be a $W^{1,p}$ connection defined on vector bundle E over a unit ball B . For some $4 > p > 2$, if*

$$\int_B |F|^2 dv < \varepsilon_0,$$

then we can find a trivialization of E in which $D = d + A$ satisfying

1. $d^*A = 0$;
2. $\|A\|_{W^{1,p}(B)} \leq C(n) \|F\|_{L^p(B)}$;
3. $\|\tilde{A}\|_{W^{1,2}(B)} \leq C(n) \|F\|_{L^p(B)}$.

2 Compactness Theorem

3 Blow-up Analysis

In this section, we discuss a sequence of Yang-Mills connection D_i defined on E over M with uniformly bounded energy. First introduce the ε -regularity theorem.

Theorem 2.2. *If D is a Yang-Mills connection on B_1 and*

$$\int_{B_1} |F|^2 \leq \varepsilon_0,$$

then

$$\sup_{B^{1/2}} |\nabla^k F| \leq C(k).$$

Proof.

□

4 Curvature Estimate for Yang-Mills Connection

In this section, we first establish a monotonicity formula for Yang-Mills connections. We then use this monotonicity formula to prove a curvature estimate for Yang-Mills connections. This theorem transforms the local L^2 control of the curvature into pointwise estimates, playing a crucial role in analyzing the regularity, singularity structure, compactness of moduli spaces, and energy distribution of Yang-Mills connections. It is worth noting that this theorem does not restrict the dimension of the base manifold to be equal to 4.

Part I: a monotonicity formula for Yang-Mills connections

We first discuss a monotonicity formula for Yang-Mills connections, which is essentially due to Price in his paper, “*A monotonicity formula for Yang-Mills fields*”. More explicitly, we will analyze the monotonicity properties of a weighted energy functional

$$\mathcal{E}(\tau) := \tau^{4-n} e^{a\tau^2} \int_{B_\tau(p)} |F_A|^2 dV_g.$$

As before, M denotes a Riemannian manifold with a metric g and E is a vector bundle over M with compact structure group G . For any connection A of E , its curvature form F_A takes values in $\mathfrak{g}$ (notice that $F_A \in \Omega^2(M, \text{Ad}(E))$ and fibers of $\text{Ad}(E)$ are Lie algebra $\mathfrak{g}$). The norm of F_A at any $p \in M$ is given by

$$|F_A|^2 = \sum_{i,j=1}^n \langle F_A(e_i, e_j), F_A(e_i, e_j) \rangle,$$

where $\{e_i\}$ is any orthonormal basis of $T_p M$, and $\langle \cdot, \cdot \rangle$ is the Killing form of $\mathfrak{g}$.

Let $\{\phi_t\}_{|t|<\infty}$ be a one-parameter family of diffeomorphisms of M , then for arbitrary fixed connection A , we can define a family of connections $\phi_t^*(A)$ as follows: denote by D the induced covariant derivative of A , τ_t^0 the parallel transport of E associated to A along the path $\phi_s(x)_{0 \leq s \leq t}$, where $x \in M$. More precisely, for any $u \in E_x$ over $x \in M$, let $\tau_s^0(u)$ be the section of E over the path $\phi_s(x)_{0 \leq s \leq t}$ such that

$$D_{\frac{\partial}{\partial s}} \tau_s^0(u) = 0, \quad \tau_0^0(u) = u.$$

We define $A^t = \phi_t^*(A)$ by defining its associated covariant derivative: for any $X \in TM$, $v \in \Gamma(M, E)$,

$$D_X^t v = (\tau_t^0)^{-1} (D_{d\phi_t(X)}(\tau_t^0(v)))$$

An natural question is: Can above definition ensure A^t be a connection? Let $Y = d\phi_t(X)$, $g(y) = f(\phi_t^{-1}(y)) = ((\phi_t^{-1})^* f)(y)$ and $S(y) = \tau_t^0(v(\phi_t^{-1}(y)))$, we apply the Leibniz

rule for the connection D to compute

$$\begin{aligned}
 D_X^t(fv)(x) &= (\tau_t^0|_{\phi_t(x)})^{-1} [D_{d\phi_t(X)} (\tau_t^0((fv)(\phi_t^{-1}(y))))]_{y=\phi_t(x)} \\
 &= (\tau_t^0|_{\phi_t(x)})^{-1} [D_{d\phi_t(X)} (f(\phi_t^{-1}(y))\tau_t^0(v(\phi_t^{-1}(y))))]_{y=\phi_t(x)} \\
 &= (\tau_t^0|_{\phi_t(x)})^{-1} [(Y(g)|_{\phi_t(x)})S(\phi_t(x)) + g(\phi_t(x))(D_{Y|_{\phi_t(x)}}S)] \\
 &= (\tau_t^0|_{\phi_t(x)})^{-1} [X(f)(x)\tau_t^0(v(x)) + f(x)D_{Y|_{\phi_t(x)}}S] \\
 &= X(f)(x)(\tau_t^0|_{\phi_t(x)})^{-1}(\tau_t^0(v(x))) + f(x)(\tau_t^0|_{\phi_t(x)})^{-1}(D_{Y|_{\phi_t(x)}}S) \\
 &= X(f)(x)v(x) + f(x)D_X^t v(x),
 \end{aligned}$$

Hence the Leibniz rule for D^t holds.

The curvature form of A^t is then given by

$$F_{A^t}(X, Y) = (\tau_t^0)^{-1} \cdot F_A(d\phi_t(X), d\phi_t(Y)) \cdot \tau_t^0. \quad (2.1)$$

It follows that (notice that when connection is compatible with metric, the parallel transport is a isometric map.)

$$\begin{aligned}
 YM(A^t) &= \frac{1}{4\pi^2} \int_M |F_{A^t}|^2 dV_g \\
 &= \frac{1}{4\pi^2} \int_M \sum_{i,j=1}^n |F_A(d\phi_t(e_i), d\phi_t(e_j))|^2 (\phi_t(x)) dV_g(x) \\
 &= \frac{1}{4\pi^2} \int_M \sum_{i,j=1}^n |F_A(d\phi_t(e_i(\phi_t^{-1}(x))), d\phi_t(e_j(\phi_t^{-1}(x))))|^2 \text{Jac}(\phi_t^{-1}) dV_g,
 \end{aligned} \quad (2.2)$$

where V_g is the volume form of metric g , and $\{e_i\}$ is any local orthonormal basis of TM .

The required monotonicity will follow from a variational formula, which will be proved in the following lemma:

Lemma 2.3. *Suppose A is a Yang-Mills connection, then*

$$0 = \int_M \left(|F_A|^2 \text{div } X - 4 \sum_{i,j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \right) dV_g. \quad (2.3)$$

Proof. We first compute the variational formula for the YM functional with respect to A^t .

Denote $K_t(x), J_t(x)$ by

$$\begin{aligned}
 K_t(x) &:= \sum_{i,j=1}^n |F_A(d\phi_t(e_i(\phi_t^{-1}(x))), d\phi_t(e_j(\phi_t^{-1}(x))))|^2; \\
 J_t(x) &:= \text{Jac}(\phi_t^{-1}).
 \end{aligned}$$

Then compute directly,

$$\begin{aligned}
 \left. \frac{d}{dt} \right|_{t=0} YM(A^t) &= \frac{1}{4\pi^2} \int_M \left. \frac{d}{dt} \right|_{t=0} [K_t(x)J_t(x)] dV_g(x) \\
 &= \frac{1}{4\pi^2} \int_M \left(\left. \frac{dK_t}{dt} \right|_{t=0} \right) J_0(x) + K_0(x) \left(\left. \frac{dJ_t}{dt} \right|_{t=0} \right) dV_g \\
 &= \frac{1}{4\pi^2} \int_M \left(\left. \frac{dK_t}{dt} \right|_{t=0} \right) + |F_A(x)|^2 \left(\left. \frac{dJ_t}{dt} \right|_{t=0} \right) dV_g.
 \end{aligned} \tag{2.4}$$

We now proceed to calculate derivative of the Jacobian $\left. \frac{dJ_t}{dt} \right|_{t=0}$ and derivative of the Curvature Term $\left. \frac{dK_t}{dt} \right|_{t=0}$ respectively.

- Computation for $\left. \frac{dJ_t}{dt} \right|_{t=0}$:

Let X be the vector generated by ϕ_t , i.e. $X(p) = \left. \frac{d}{dt} \right|_{t=0} \phi_t(p)$, then $\left. \frac{d}{dt} \right|_{t=0} \phi_t^{-1}(p) = -X$. We compute $\left. \frac{dJ_t}{dt} \right|_{t=0}$:

$$\begin{aligned}
 \left. \frac{dJ_t}{dt} \right|_{t=0} &= \left. \frac{d}{dt} \right|_{t=0} \det(d\phi_t^{-1}) \\
 &= \det(d\phi_0^{-1}) \text{Tr} \left((d\phi_0^{-1})^{-1} \left. \frac{d}{dt} \right|_{t=0} (d\phi_t^{-1}) \right) \\
 &= \text{Tr} \left(d \left(\left. \frac{d}{dt} \right|_{t=0} \phi_t^{-1} \right) \right) \\
 &= \text{Tr}(d(-X)) \\
 &= -\text{div} X
 \end{aligned}$$

- Computation for $\left. \frac{dK_t}{dt} \right|_{t=0}$:

Denote $u_i(t, x)$ by $d\phi_t(e_i(\phi_t^{-1}(x)))$, then $e_i = u_i|_{t=0}$. We use an identity $v_i := \left. \frac{du_i}{dt} \right|_{t=0} = -[X, e_i]$ in the following computation, which can be verified directly via the definition of Lie derivative.

$$\begin{aligned}
 \left. \frac{d}{dt} \right|_{t=0} K_t &= \sum_{i,j} 2 \left\langle \left. \frac{d}{dt} \right|_{t=0} F_A(u_i(t, x), u_j(t, x)), F_A(u_i(0, x), u_j(0, x)) \right\rangle \\
 &= \sum_{i,j} 2 \left\langle F_A \left(\left. \frac{du_i}{dt} \right|_{t=0}, u_j(0) \right) + F_A \left(u_i(0), \left. \frac{du_j}{dt} \right|_{t=0} \right), F_A(u_i(0, x), u_j(0, x)) \right\rangle \\
 &= \sum_{i,j} 2 \langle F_A(v_i, e_j) + F_A(e_i, v_j), F_A(e_i, e_j) \rangle \\
 &= 4 \sum_{i,j} \langle F_A(v_i, e_j), F_A(e_i, e_j) \rangle \\
 &= -4 \sum_{i,j} \langle F_A([X, e_i], e_j), F_A(e_i, e_j) \rangle.
 \end{aligned}$$

Substitute $\frac{dK_t}{dt}|_{t=0}$ and $\frac{dJ_t}{dt}|_{t=0} = -\operatorname{div} X$ into (2.4) yields

$$\left. \frac{d}{dt} \right|_{t=0} YM(A^t) = -\frac{1}{4\pi^2} \int_M 4 \sum_{i,j} \langle F_A([X, e_i], e_j), F_A(e_i, e_j) \rangle + |F_A(x)|^2 \operatorname{div} X dV_g. \quad (2.5)$$

Since $[X, e_i] = \nabla_X e_i - \nabla_{e_i} X$, where ∇ denotes the Levi-Civita connection of g , we obtain

$$\begin{aligned} \sum_{i,j=1}^n \langle F_A([X, e_i], e_j), F_A(e_i, e_j) \rangle &= - \sum_{i,j=1}^n (\langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle - \langle F_A(\nabla_X e_i, e_j), F_A(e_i, e_j) \rangle) \\ &= - \sum_{i,j=1}^n (\langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \\ &\quad - g(\nabla_X e_i, e_k) \langle F_A(e_k, e_j), F_A(e_i, e_j) \rangle) \\ &= - \sum_{i,j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \end{aligned}$$

where in the last step, we used $g(\nabla_X e_i, e_k) = -g(e_i, \nabla_X e_k) = -g(\nabla_X e_k, e_i)$. Combine this computation with (2.5) and the fact that A is Yang-Mills connection, we obtain (2.3). $\square$

Now we derive a monotonicity formula via above variational lemma.

Fix any $p \in M$, let r_p be a positive number with properties: there are normal coordinates $x_1, \dots, x_n$ in the geodesic ball $B_{r_p}(p)$ of (M, g) , such that $p = (0, \dots, 0)$ and for some constant $C(p)$,

$$\begin{aligned} |g_{ij} - \delta_{ij}| &\leq C(p)(|x_1|^2 + \dots + |x_n|^2), \\ |dg_{ij}| &\leq C(p)\sqrt{|x_1|^2 + \dots + |x_n|^2}. \end{aligned}$$

The constants r_p and $C(p)$ can be chosen depending only on the injective radius at p and the curvature of g . If $M = \mathbb{R}^n$ and g is flat, we can take $r_p = \infty$ and $C(p) = 0$.

Let $r(x)$ be the distance function from p ; i.e. $r(x) = \sqrt{x_1^2 + \dots + x_n^2}$, ϕ be a positive function on the unit sphere S^{n-1} , define a vector field

$$X(x) = \xi(r)\phi\left(\frac{x}{r}\right)r\frac{\partial}{\partial r} = \xi(r)\phi\left(\frac{x}{r}\right)\left(\sum_i x_i \frac{\partial}{\partial x_i}\right),$$

where ξ is some smooth function with compact support in $B_{r_p}(p)$.

Let $\{e_1, \dots, e_n\}$ be any orthonormal basis near p such that $e_1 = \frac{\partial}{\partial r}$. Since $x_1, \dots, x_n$ are normal coordinates, we have $\nabla_{\frac{\partial}{\partial r}} \frac{\partial}{\partial r} = 0$, thus

$$\begin{aligned} \nabla_{\frac{\partial}{\partial r}} X &= \nabla_{\frac{\partial}{\partial r}} \left(\xi(r)\phi(\theta)r\frac{\partial}{\partial r} \right) \\ &= \frac{\partial}{\partial r} (\xi(r)\phi(\theta)r) \frac{\partial}{\partial r} \\ &= (\xi' r + \xi)\phi(\theta) \frac{\partial}{\partial r}, \end{aligned} \quad (2.6)$$

where $\theta = \frac{x}{r}$. Moreover, for $i \geq 2$,

$$\begin{aligned}
 \nabla_{e_i} X &= \nabla_{e_i} \left(\xi(r) \phi(\theta) r \frac{\partial}{\partial r} \right) \\
 &= e_i (\xi \phi(\theta) r) \frac{\partial}{\partial r} + \xi \phi(\theta) r \nabla_{e_i} \frac{\partial}{\partial r} \\
 &= \xi r e_i (\phi(\theta)) \frac{\partial}{\partial r} + \xi \phi(\theta) \sum_{j=1}^n b_{ij} e_j.
 \end{aligned} \tag{2.7}$$

One can readily check that for the coefficients b_{ij} , we have asymptotic estimates (we always denote by $O(1)$ a quantity bounded by a constant depending only on n .)

$$|b_{ij} - \delta_{ij}| = O(1)C(p)r^2.$$

Actually, let $c_{ij} = g(\nabla_{e_i} \frac{\partial}{\partial r}, e_j)$ be the components of $\nabla_{e_i} \frac{\partial}{\partial r}$ in the orthonormal basis $\{e_j\}$, then $b_{ij} = r c_{ij}$. Clearly, $c_{i1} = c_{1j} = 0$ and $c_{ij} = II(e_i, e_j)$ are the components of the second fundamental form of the geodesic sphere S_r . Moreover, $c_{ij} = \frac{1}{r} \delta_{ij}$ in flat space and $c_{ij} \approx \frac{1}{r} \delta_{ij} - \frac{r}{3} R_{1i1j} + O(r^2)$ in curved space, thus for b_{ij} , we have

1. $b_{i1} = b_{1j} = 0$.
2. For $i, j \geq 2$, $b_{ij} = \delta_{ij}$ in flat space and $b_{ij} \approx \delta_{ij} - \frac{r^2}{3} R_{1i1j} + O(r^3)$ in curved space.

We now utilize equations (2.6) and (2.7) in conjunction with the first variational formula (2.3) to derive a new equation (2.4.13).

Rewrite equation (2.3) as

$$\int_M |F_A|^2 \operatorname{div} X \, dV_g = 4 \int_M \sum_{i,j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle dV_g. \tag{2.8}$$

Observe that

$$\begin{aligned}
 \operatorname{div} X &= \sum_{i=1}^n g(\nabla_{e_i} X, e_i) \\
 &= (\xi' r + \xi) \phi + \xi \phi \sum_{i=2}^n b_{ii} \\
 &= (\xi' r + n\xi) \phi + O(1)C(p)r^2 \xi \phi.
 \end{aligned} \tag{2.9}$$

Regarding $S = 4 \sum_{i,j} \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle$, we split this into two parts:

- Case $i = 1$:

$$S_1 = 4 \sum_{j=1}^n \langle F_A(\nabla_{e_1} X, e_j), F_A(e_1, e_j) \rangle,$$

- Case $i \geq 2$:

$$S_2 = 4 \sum_{i=2}^n \sum_{j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle.$$

The computation of S_1 is straightforward,

$$\begin{aligned} S_1 &= 4 \sum_{j=1}^n \langle F_A(\nabla_{e_1} X, e_j), F_A(e_1, e_j) \rangle \\ &= 4(\xi' r + \xi) \phi \sum_{j=2}^n |F_A(e_1, e_j)|^2 \\ &= 4(\xi' r + \xi) \phi |\iota_{e_1} F_A|^2. \end{aligned} \tag{2.10}$$

Split S_2 into (notice that $e_1(\phi) = 0$)

$$\begin{aligned} S_{2a} &= 4\xi r \sum_{i=2}^n \sum_{j=1}^n e_i(\phi) \langle F_A(e_1, e_j), F_A(e_i, e_j) \rangle \\ &= 4\xi r \sum_{j=1}^n \left(\langle F_A(e_1, e_j), \sum_{i=1}^n e_i(\phi) F_A(e_i, e_j) \rangle \right) \\ &= 4\xi r \langle \iota_{e_1} F_A, \iota_{\nabla \phi} F_A \rangle \end{aligned}$$

and

$$\begin{aligned} S_{2b} &= 4\xi \phi \sum_{i=2}^n \sum_{j=1}^n \sum_{l=1}^n b_{il} \langle F_A(e_l, e_j), F_A(e_i, e_j) \rangle \\ &= 4\xi \phi \sum_{i=2}^n \sum_{j=1}^n \sum_{l=1}^n (\delta_{il} + O(1)C(p)r^2) \langle F_A(e_l, e_j), F_A(e_i, e_j) \rangle \\ &:= S_{2b,\delta} + S_{2b,O(r^2)}, \end{aligned}$$

where the leading term $S_{2b,\delta}$ and the residual term $S_{2b,O(r^2)}$ are

$$\begin{aligned} S_{2b,\delta} &= 4\xi \phi \sum_{i=2}^n \sum_{j=1}^n \langle F_A(e_i, e_j), F_A(e_i, e_j) \rangle; \\ S_{2b,O(r^2)} &= 4\xi \phi \sum_{i=2}^n \sum_{j=1}^n \sum_{l=1}^n O(1)C(p)r^2 \langle F_A(e_l, e_j), F_A(e_i, e_j) \rangle. \end{aligned}$$

For the leading term $S_{2b,\delta}$,

$$\begin{aligned} 4\xi \phi \sum_{i=2}^n \sum_{j=1}^n \langle F_A(e_i, e_j), F_A(e_i, e_j) \rangle &= 4\xi \phi \sum_{i=2}^n |F_A(e_i, \cdot)|^2 \\ &= 4\xi \phi \left(\sum_{i=1}^n |F_A(e_i, \cdot)|^2 - |F_A(e_1, \cdot)|^2 \right) \\ &= 4\xi \phi (|F_A|^2 - |\iota_{e_1} F_A|^2) \end{aligned}$$

Hence

$$S_2 = 4\xi r \langle \iota_{e_1} F_A, \iota_{\nabla \phi} F_A \rangle + 4\xi \phi (|F_A|^2 - |\iota_{e_1} F_A|^2) + \xi \phi O(1)C(p)r^2. \quad (2.4.11)$$

Substitute (2.4.9), (2.4.10) and (2.4.11) into the main identity (2.4.8), we obtain

$$\begin{aligned} \int_M |F_A|^2 (\xi' r + n\xi + O(1)C(p)r^2) \phi dV_g &= \int_M 4(\xi' r + \xi) \phi |\iota_{e_1} F_A|^2 \\ &\quad + 4\xi r \langle \iota_{e_1} F_A, \iota_{\nabla \phi} F_A \rangle + 4\xi \phi (|F_A|^2 - |\iota_{e_1} F_A|^2) dV_g. \end{aligned}$$

Simplify this identity yields

$$\int_M |F_A|^2 (\xi' r + (n-4)\xi + O(1)C(p)r^2\xi) \phi dV_g = 4 \int_M \left(\xi' r \phi \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 + \xi r \langle \iota_{\frac{\partial}{\partial r}} F_A, \iota_{\nabla \phi} F_A \rangle \right) dV_g. \quad (2.4.13)$$

We turn to specific selections for the functions ϕ and ξ to analyze identity (2.4.13). For any τ small enough, put $\xi(r) = \xi_\tau(r) = \eta(r/\tau)$, where η is smooth and satisfies: $\eta(r) = 1$ for $r \in [0, 1]$, $\eta(r) = 0$ for $r \in [1 + \varepsilon, \infty)$, $\varepsilon > 0$ and $\eta'(r) \leq 0$. Then $\tau \frac{\partial}{\partial \tau}(\xi_\tau(r)) = -r\xi'_\tau(r)$, plug this into (2.4.13), we obtain (for $\tau \geq r$)

$$\begin{aligned} &\tau \frac{\partial}{\partial \tau} \left(\int_M \xi_\tau \phi |F_A|^2 dV_g \right) + ((4-n) + O(1)C(p)\tau^2) \int_M \xi_\tau \phi |F_A|^2 dV_g \\ &= 4\tau \frac{\partial}{\partial \tau} \left(\int_M \xi_\tau \phi \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 dV_g \right) - 4 \int_M \xi_\tau r \langle \iota_{\frac{\partial}{\partial r}} F_A, \iota_{\nabla \phi} F_A \rangle dV_g. \end{aligned} \quad (2.4.14)$$

We adopt the following notation for simplicity:

$$\begin{aligned} I(\tau) &= \int_M \xi_\tau \phi |F_A|^2 dV_g; \\ J(\tau) &= \int_M \xi_\tau \phi \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 dV_g; \\ Q(\tau) &= \int_M \xi_\tau r \langle \iota_{\frac{\partial}{\partial r}} F_A, \iota_{\nabla \phi} F_A \rangle dV_g; \\ \epsilon(\tau) &= O(1)C(p)\tau^2, \end{aligned}$$

then the starting equation (2.4.14) reads

$$\tau \frac{dI}{d\tau} + (4-n)I(\tau) + \epsilon(\tau)I(\tau) = 4\tau \frac{dJ}{d\tau} - 4Q(\tau). \quad (2.4.15)$$

Let $L(\tau) = \tau^{4-n} e^{\pm a\tau^2} I(\tau)$. We compute $\frac{dL}{d\tau}$ using the product rule:

$$\begin{aligned} \frac{dL}{d\tau} &= ((4-n)\tau^{3-n}) e^{\pm a\tau^2} I(\tau) + \tau^{4-n} \left(e^{\pm a\tau^2} (\pm 2a\tau) \right) I(\tau) + \tau^{4-n} e^{\pm a\tau^2} \frac{dI}{d\tau} \\ &= \tau^{4-n} e^{\pm a\tau^2} \left(\frac{1}{\tau} \left((4-n)I(\tau) + (\pm 2a\tau^2)I(\tau) + \tau \frac{dI}{d\tau} \right) \right). \end{aligned}$$

From (2.4.15), we solve for $\tau \frac{dI}{d\tau}$

$$\tau \frac{dI}{d\tau} = 4\tau \frac{dJ}{d\tau} - 4Q(\tau) - (4-n)I(\tau) - \epsilon(\tau)I(\tau),$$

substitute this into the expression for the $\frac{dL}{d\tau}$ and compute

$$\begin{aligned} \frac{dL}{d\tau} &= \tau^{4-n} e^{\pm a\tau^2} \left(\frac{1}{\tau} \left((\pm 2a\tau^2)I(\tau) + 4\tau \frac{dJ}{d\tau} - 4Q(\tau) - \epsilon(\tau)I(\tau) \right) \right) \\ &= 4\tau^{4-n} e^{\pm a\tau^2} \left(\frac{dJ}{d\tau} - \tau^{-1}Q(\tau) + \left(\pm \frac{a}{2} - O(1)C(p) \right) \tau I(\tau) \right). \end{aligned}$$

Choose the nonnegative constant $a \geq \max\{2O(1)C(p), -2O(1)C(p)\}$, we deduce from the above

$$\begin{aligned} &\frac{\partial}{\partial \tau} \left(\tau^{4-n} e^{\pm a\tau^2} \int_M \xi_\tau \phi |F_A|^2 dV_g \right) \\ &= 4\tau^{4-n} e^{\pm a\tau^2} \left(\frac{\partial}{\partial \tau} \left(\int_M \xi_\tau \phi \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 dV_g \right) \right. \\ &\quad \left. + \left(\pm \frac{a}{2} - O(1)C(p) \right) \tau \int_M \xi_\tau \phi |F_A|^2 dV_g - \tau^{-1} \int_M \xi_\tau \langle \iota_{\frac{\partial}{\partial r}} F_A, \iota_{\nabla \phi} F_A \rangle dV_g \right). \end{aligned} \quad (2.11)$$

As for the upper sign (+), the coefficient of $\tau I(\tau)$ is $(a/2 - O(1)C(p))\tau \geq 0$. Since $I(\tau) \geq 0$, the second term in RHS of (2.11) is non-negative, thus:

$$\frac{d}{d\tau} L(\tau) \geq 4\tau^{4-n} e^{+a\tau^2} \left(\frac{dJ}{d\tau} - \tau^{-1}Q(\tau) \right).$$

As for the lower sign (-), the coefficient of $\tau I(\tau)$ is $(-a/2 - O(1)C(p))\tau \leq 0$. Since $I(\tau) \geq 0$, the second term in RHS of (2.11) is non-positive, thus

$$\frac{d}{d\tau} L(\tau) \leq 4\tau^{4-n} e^{-a\tau^2} \left(\frac{dJ}{d\tau} - \tau^{-1}Q(\tau) \right).$$

We can combine these two cases as:

$$\pm \frac{d}{d\tau} L(\tau) \geq \pm 4\tau^{4-n} e^{\pm a\tau^2} \left(\frac{dJ}{d\tau} - \tau^{-1}Q(\tau) \right)$$

Integrate the inequality from σ to ρ (where $0 < \sigma < \rho$):

$$\begin{aligned} \pm(L(\rho) - L(\sigma)) &= \int_\sigma^\rho \pm \frac{d}{d\tau} L(\tau) d\tau \\ &\geq \pm 4 \int_\sigma^\rho \tau^{4-n} e^{\pm a\tau^2} \frac{dJ}{d\tau} d\tau \mp 4 \int_\sigma^\rho \tau^{3-n} e^{\pm a\tau^2} Q(\tau) d\tau. \end{aligned} \quad (2.4.17)$$

Recall $Q(\tau) = \int_M \xi_\tau r \langle \iota_{e_1} F_A, \iota_{\nabla \phi} F_A \rangle dV_g$ (with $e_1 = \partial/\partial r$), use the Cauchy-Schwarz inequality $|\langle u, v \rangle| \leq |u||v|$, we have $\langle \iota_{e_1} F_A, \iota_{\nabla \phi} F_A \rangle \geq -|\iota_{e_1} F_A| |\iota_{\nabla \phi} F_A|$, therefore

$$\mp Q(\tau) \geq - \int_M \xi_\tau r |\iota_{e_1} F_A| |\iota_{\nabla \phi} F_A| dV_g.$$

Substitute this into the inequality (2.4.17), we obtain

$$\begin{aligned} \pm(L(\rho) - L(\sigma)) &\geq \pm 4 \int_{\sigma}^{\rho} \tau^{4-n} e^{\pm a\tau^2} \frac{dJ}{d\tau} d\tau \\ &\quad - 4 \int_{\sigma}^{\rho} \tau^{3-n} e^{\pm a\tau^2} \left(\int_M \xi_{\tau} r |\iota_{e_1} F_A| |\iota_{\nabla\phi} F_A| dV_g \right) d\tau. \end{aligned} \quad (2.4.18)$$

Taking $\varepsilon \rightarrow 0$ in inequality (2.4.18), we study the limiting behavior and determine the resulting conclusions. As $\varepsilon \rightarrow 0$ in the cutoff function $\xi_{\tau} = \eta(r/\tau)$ implies $\xi_{\tau} \rightarrow \chi_{B_{\tau}(p)}$, the integrals become integrals over the ball $B_{\tau}(p)$, i.e.

- $L(\tau) \rightarrow \tau^{4-n} e^{\pm a\tau^2} \int_{B_{\tau}(p)} \phi |F_A|^2 dV_g$,
- $\int_M \xi_{\tau} r |\iota_{e_1} F_A| |\iota_{\nabla\phi} F_A| dV_g \rightarrow \int_{B_{\tau}(p)} r |\iota_{e_1} F_A| |\iota_{\nabla\phi} F_A| dV_g$.

For the J term, we claim a fact involving $dJ/d\tau$:

$$\lim_{\varepsilon \rightarrow 0} \frac{dJ}{d\tau} = \int_{\partial B_{\tau}(p)} \phi |\iota_{e_1} F_A|^2 dS_{\tau}. \quad (2.4.19)$$

proof of the claim. Let's calculate $\frac{dJ}{d\tau}$ precisely using the definition $J(\tau) = \int_M \xi_{\tau}(r) f(x) dV_g$, where $f(x) = \phi(x) |\iota_{e_1} F_A|^2(x)$. Denote $H(r) = \int_{\partial B_r(p)} f(y) dS_r(y)$ (here $H(r)$ does not depend on τ), then

$$J(\tau) = \int_0^{\infty} \eta(r/\tau) H(r) dr$$

and

$$\frac{dJ}{d\tau} = \int_0^{\infty} \frac{\partial}{\partial \tau} (\eta(r/\tau)) H(r) dr = -\frac{1}{\tau^2} \int_0^{\infty} r \eta'(r/\tau) H(r) dr$$

Now, consider the limit $\varepsilon \rightarrow 0$. Obviously, the smooth function $\eta(s)$ approaches the Heaviside step function $H(1-s)$ (1 for $s \leq 1$, 0 for $s > 1$) as $\varepsilon \rightarrow 0$. Correspondingly, its derivative $\eta'(s)$ approaches the negative Dirac delta function centered at $s = 1$: $\eta'(s) \rightarrow -\delta(s-1)$. In this limit, the expression for $\frac{dJ}{d\tau}$ becomes:

$$\lim_{\varepsilon \rightarrow 0} \frac{dJ}{d\tau} = -\frac{1}{\tau^2} \int_0^{\infty} r \left(-\delta\left(\frac{r}{\tau} - 1\right) \right) H(r) dr.$$

Using the property $\delta(ax-b) = \frac{1}{|a|} \delta(x-b/a)$, we have $\delta(r/\tau - 1) = \tau \delta(r - \tau)$, thus

$$\lim_{\varepsilon \rightarrow 0} \frac{dJ}{d\tau} = \frac{1}{\tau} \int_0^{\infty} r \delta(r - \tau) H(r) dr = \frac{1}{\tau} (\tau H(\tau)) = H(\tau) = \int_{\partial B_{\tau}(p)} \phi |\iota_{e_1} F_A|^2 dS_{\tau}.$$

That is (2.4.19). Thanks to (2.4.19), we have

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \int_{\sigma}^{\rho} \tau^{4-n} e^{\pm a\tau^2} \frac{dJ}{d\tau} d\tau &= \int_{\sigma}^{\rho} \tau^{4-n} e^{\pm a\tau^2} \left(\int_{\partial B_{\tau}(p)} \phi |\iota_{e_1} F_A|^2 dS_{\tau} \right) d\tau \\ &= \int_{B_{\rho}(p) \setminus B_{\sigma}(p)} r^{4-n} e^{\pm ar^2} \phi |\iota_{e_1} F_A|^2 dV_g. \end{aligned}$$

Substitute all the limits we have concluded above into the inequality (2.4.18), we obtain

$$\begin{aligned}
 & \pm \left(\rho^{4-n} e^{\pm a \rho^2} \int_{B_\rho(p)} \phi |F_A|^2 dV_g - \sigma^{4-n} e^{\pm a \sigma^2} \int_{B_\sigma(p)} \phi |F_A|^2 dV_g \right) \\
 & \geq \pm 4 \int_{B_\rho(p) \setminus B_\sigma(p)} r^{4-n} e^{\pm a r^2} \phi |\iota_{e_1} F_A|^2 dV_g \\
 & \quad - 4 \int_\sigma^\rho \tau^{3-n} e^{\pm a \tau^2} \left(\int_{B_\tau(p)} |\iota_{e_1} F_A| |\iota_{\nabla \phi} F_A| dV_g \right) d\tau.
 \end{aligned} \tag{2.4.20}$$

Take $\phi = 1$, (2.4.20) yields

Theorem 2.4 (Price). *Let r_p , $C(p)$ and a be as above. Then for any $0 < \sigma < \rho < r_p$, we have*

$$\rho^{4-n} e^{a \rho^2} \int_{B_\rho(p)} |F_A|^2 dV_g - \sigma^{4-n} e^{a \sigma^2} \int_{B_\sigma(p)} |F_A|^2 dV_g \geq 4 \int_{B_\rho(p) \setminus B_\sigma(p)} r^{4-n} e^{a r^2} \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 dV_g. \tag{2.4.21}$$

Moreover, if $M = \mathbb{R}^n$ and g is flat, then the equality holds in (2.4.21) for $\rho \in (0, \infty)$ and $a = 0$.

Remark 1. Both (2.4.20) and (2.4.21) still hold when A is only a smooth Yang-Mills connection on $M \setminus \{p\}$ with $\int_M |F_A|^2 dV_g < \infty$. To see this, we replace η in defining ξ by η_ε for sufficiently small ε , where $\eta_\varepsilon(t) = 0$ for either $t \leq \varepsilon$ or $t \geq 1 + \varepsilon$, and $\eta_\varepsilon(t) = 1$ for $t \in (\varepsilon, 1 - \varepsilon)$. Then we can follow the same arguments and obtain both (2.4.20) and (2.4.21) for such a Yang-Mills connection A with isolated singularity at p .

Remark 2. It is easy to see from Theorem (2.4.2) that the weighted energy functional $\mathcal{E}(\tau)$ is strictly increasing with respect to τ , and inequality (2.4.21) provides a lower bound estimate for the difference of $\mathcal{E}(\tau)$.

Part II: a curvature estimate for Yang-Mills connections

Now we give a basic curvature for Yang-Mills connection. This estimate was first derived by Uhlenbeck.

Theorem 2.5 (Uhlenbeck). *Let A be any Yang-Mills connection of a G -bundle E over M . Then there are $\varepsilon = \varepsilon(n) > 0$ and $C = C(n) > 0$, which depend only on n and M , such that for any $p \in M$ and $\rho < r_p$, whenever*

$$\rho^{4-n} \int_{B_\rho(p)} |F_A|^2 dV_g \leq \varepsilon,$$

then

$$|F_A|(p) \leq \frac{C}{\rho^2} \left(\rho^{4-n} \int_{B_\rho(p)} |F_A|^2 dV_g \right)^{\frac{1}{2}}. \tag{2.4.22}$$

Proof. Our proof here uses R. Schoen's arguments in “*Analytic aspects of the harmonic maps problem*” for harmonic maps.

Define a function

$$f(r) = (1 - 2r)^2 \sup_{x \in B_r(p)} |F_A|(x), \quad r \in [0, 1/2].$$

Then $f(r)$ is continuous in $[0, \frac{1}{2}]$ with $f(\frac{1}{2}) = 0$, so that f attains its maximum at a certain r_0 in $[0, \frac{1}{2}]$. For this maximum point, we claim that $f(r_0) \leq 64$ if ε is sufficiently small.

Assume that $f(r_0) > 64$. Put $b = \sup_{x \in B_{r_0}(p)} |F_A|(x) = |F_A|(x_0)$, take $\sigma = \frac{1}{4}(1 - 2r_0)$, then we have

$$\sup_{x \in B_\sigma(x_0)} |F_A| \leq \sup_{x \in B_{r_0+\sigma}(p)} |F_A|(x) \leq \frac{(1 - 2r_0)^2}{(1 - 2r_0 - 2\sigma)^2} \sup_{x \in B_{r_0}(p)} |F_A|(x) = 4b.$$

Define a scaled metric $\tilde{g} = bg$, then with respect to $\tilde{g}$, the norm $|F_A|_{\tilde{g}}$ of F_A is equal to $b^{-1}|F_A|_g$. Notice that the radius σ in the original metric corresponds to radius $\sqrt{b} \cdot \sigma$ in the rescaled metric $\tilde{g}$, and obviously, $16\sigma^2 b \geq 64 \Rightarrow \sigma\sqrt{b} \geq 2$, hence $B_2(x_0, \tilde{g}) \subset B_{\sigma\sqrt{b}}(x_0, \tilde{g}) = B_\sigma(x_0, g)$. Therefore,

$$\sup_{B_2(x_0, \tilde{g})} |F_A|_{\tilde{g}} \leq \sup_{B_{\sigma\sqrt{b}}(x_0, \tilde{g})} |F_A|_{\tilde{g}} \leq 4. \quad (2.4.23)$$

Since A is a Yang-Mills connection, by the second Bianchi identity and straightforward computations, we can derive the following equation:

$$\frac{1}{2}\Delta_{\tilde{g}}|F_A|_{\tilde{g}}^2 = |\tilde{\nabla}F_A|_{\tilde{g}}^2 - 2F_A \# F_A \# R(\tilde{g}) - 2F_A * F_A * F_A, \quad (2.4.24)$$

where $F_A \# F_A \# R(\tilde{g})$ and $F_A * F_A * F_A$ are defined as follows: in any orthonormal basis $\{e_1, \dots, e_n\}$ of $\tilde{g}$,

$$\begin{aligned} F_A \# F_A \# R(\tilde{g}) &= \sum_{l,k,i,j} (\langle F_A(e_l, e_k), F_A(e_i, e_j) \rangle \\ &\quad - \sum_m \langle F_A(e_l, e_m), F_A(e_i, e_m) \rangle \delta_{jk}) R(\tilde{g})(e_l, e_j, e_k, e_i), \end{aligned}$$

and

$$F_A * F_A * F_A = \sum_{i,j,k} \langle [F_A(e_i, e_j), F_A(e_j, e_k)], F_A(e_k, e_i) \rangle.$$

Actually, there are uniform constants C_1, C_2 , depending only on n , such that

$$-\Delta_{\tilde{g}}|F_A|_{\tilde{g}} \leq C_1|F_A|_{\tilde{g}} + C_2|F_A|_{\tilde{g}}^2. \quad (2.4.25)$$

Using (2.4.23), we deduce from (2.4.25) that in $B_2(x_0, \tilde{g})$,

$$-\Delta_{\tilde{g}}|F_A|_{\tilde{g}} \leq (C_1 + 4C_2)|F_A|_{\tilde{g}}.$$

The standard Moser Iteration reads

$$1 = |F_A|_{\tilde{g}}(x_0) \leq C_3 \left(\int_{B_1(x_0, \tilde{g})} |F_A|_{\tilde{g}}^2 dV_{\tilde{g}} \right)^{\frac{1}{2}}, \quad (2.4.26)$$

where C_3 is some uniform constant. However, by the monotonicity formula (Theorem 2.4.2),

$$\begin{aligned} \int_{B_1(x_0, \tilde{g})} |F_A|_{\tilde{g}}^2 dV_{\tilde{g}} &= (\sqrt{b})^{n-4} \int_{B_{\frac{1}{\sqrt{b}}}(x_0, g)} |F_A|^2 dV_g \\ &\leq \left(\frac{1}{2}\right)^{4-n} e^{\frac{a}{4} - \frac{a}{b}} \int_{B_{\frac{1}{2}}(x_0, g)} |F_A|^2 dV_g \\ &\leq e^{\frac{a}{4} - \frac{a}{b}} \varepsilon. \end{aligned}$$

Combining this with (2.4.26), we obtain

$$1 \leq C_3 e^{\frac{a}{8} - \frac{a}{2b}} \sqrt{\varepsilon}.$$

It is impossible when $\varepsilon = \varepsilon(n)$ is sufficiently small. The claim is proved, i.e. $f(r_0) \leq 64$ if ε is sufficiently small.

Thus, we have

$$\sup_{x \in B_{\frac{1}{4}}(p)} |F_A|(x) \leq 4f(r_0) \leq 256.$$

It follows from this and (2.4.24) with $\tilde{g}$ replaced by g that for some uniform constant C_4 ,

$$-\Delta_g |F_A| \leq C_4 |F_A|_g. \quad (2.4.27)$$

To proceed (2.4.22), we still need standard Moser iteration based on (2.4.27) involed a scaling.

Rescale the ball $B_\rho(p) \subset M$ to unit size via the change of variables $\tilde{x} = \frac{x-p}{\rho}$, so that $\tilde{g} := \rho^{-2}g$. Under this rescaling, the metric becomes $\tilde{g}$, the Laplacian rescales as $\Delta_g = \rho^{-2}\Delta_{\tilde{g}}$, and the function $u := |F_A|_g$ transforms to $\tilde{u} := |F_A|_{\tilde{g}}$. Then the differential inequality (2.4.27) becomes

$$-\Delta_{\tilde{g}} \tilde{u} \leq C_4 \rho^2 \tilde{u},$$

in the unit ball $B_1(0, \tilde{g})$.

Apply Moser iteration to this inequality to conclude the pointwise bound

$$\tilde{u}(0) \leq C_5 \left(\int_{B_1} \tilde{u}^2 dV_{\tilde{g}} \right)^{1/2}.$$

Transforming back to the original metric g , we obtain:

$$|F_A|(p) \leq \frac{C_5}{\rho^2} \left(\rho^{4-n} \int_{B_\rho(p)} |F_A|^2 dV_g \right)^{1/2},$$

which is exactly the desired estimate (2.4.22). □

5 Curvature Estimate for Yang-Mills-Higgs Connection

Let (M^n, g) be a smooth Riemannian manifold whose dimension is at least four, G a compact Lie group, and $P \rightarrow M$ a principal G -bundle. $P \times_G V$ is a vector bundle associated to a representation $\rho : G \rightarrow \text{Aut}(V)$. Let A be a connection on P , inducing a covariant derivative D^{as} on sections of $P \times_G V$, and let $\phi \in \Gamma(P \times_G V)$ be a section (called the *Higgs field*). Let F_A denote the curvature of A , and let $V : V \rightarrow \mathbb{R}$ be a G -invariant potential function. Then the **Yang–Mills–Higgs (YMH) functional** is defined by

$$\mathcal{YM}\mathcal{H}(A, \phi) = \frac{1}{4\pi^2} \int_M (|F_A|^2 + |D^{as}\phi|^2 + V(\phi)) dV_g, \quad (2.5.1)$$

where the norms are taken using the Riemannian metric on M and the Hermitian or Euclidean structure on $\text{Ad}(P)$ and $P \times_G V$.

Throughout this work, we refer to the pair (A, ϕ) as a **Yang-Mills-Higgs connection**, where A is a connection on a principal G -bundle and ϕ is a section of the associated vector bundle. This terminology emphasizes the coupled nature of the gauge and scalar fields in the YMH functional.

Part I: a variational formula for YMH connection

We first define a natural coupled variation (A^t, ϕ^t) of the Yang-Mills-Higgs fields. Let $\{\psi_t\}_{|t|<\varepsilon}$ be a one-parameter family of diffeomorphisms of M . Let A be an arbitrary but fixed connection, which induces a covariant derivative D^{Ad} acting on $\text{Ad}(P)$, and also induces another covariant derivative D^{as} acting on $\Gamma(P \times_G V)$. Define a family of pulled-back connections $A^t = \psi_t^*(A)$ as follows: denote by $\tau_t^0 : (P \times_G V)_x \rightarrow (P \times_G V)_{\psi_t(x)}$ the parallel transport along the path $\psi_s(x)$ with respect to A , for $0 \leq s \leq t$. That is, for any $u \in E_x$,

$$D_{\frac{\partial}{\partial s}}^{Ad} \tau_s^0(u) = 0, \quad \tau_0^0(u) = u.$$

Subsequently, we define the covariant derivative associated to A^t : for any $X \in TM$, $v \in \Gamma(\text{Ad}(P))$,

$$D_X^{Ad,t} v = (\tau_t^0)^{-1} (D_{d\psi_t(X)}^{Ad} (\tau_t^0 v)).$$

One can check this satisfies the Leibniz rule and hence defines a connection.

We define the variation of the Higgs field ϕ using the parallel transport τ_t^0 along the path $\{\psi_s(x)\}_{0 \leq s \leq t}$ with respect to the original connection A :

$$\phi^t(x) := (\tau_t^0)^{-1} \cdot \phi(\psi_t(x)).$$

The covariant derivative of the perturbed Higgs field ϕ^t with respect to the perturbed connection $A^t := \psi_t^*(A)$ is defined analogously by

$$D_X^{t,as} \phi^t(x) := (\tau_t^0)^{-1} \cdot D_{d\psi_t(X)}^{as} (\phi(\psi_t(x))),$$

for any vector field $X \in \Gamma(TM)$.

It follows that (notice that when connection is compatible with metric, the parallel transport is a isometric map.)

$$\begin{aligned} \mathcal{YM}\mathcal{H}(A^t, \phi^t) &= \frac{1}{4\pi^2} \int_M (|F_{A^t}|^2 + |D^{as,t} \phi^t|^2 + V(\phi^t)) dV_g \\ &= \frac{1}{4\pi^2} \int_M \left(\sum_{i,j} |F_A(d\psi_t(e_i), d\psi_t(e_j))(\psi_t(x))|^2 \right. \\ &\quad \left. + \sum_i |D_{d\psi_t(e_i)}^{as} \phi(\psi_t(x))|^2 + V(\phi(\psi_t(x))) \right) dV_g. \\ &= \frac{1}{4\pi^2} \int_M \left(\sum_{i,j} |F_A(d\psi_t(e_i(\psi_t^{-1}(x))), d\psi_t(e_j(\psi_t^{-1}(x))))|^2 \right. \\ &\quad \left. + \sum_i |D_{d\psi_t(e_i(\psi_t^{-1}(x)))}^{as} \phi(x)|^2 + V(\phi(x)) \right) \cdot |\det(d\psi_t^{-1}(x))| dV_g. \end{aligned} \tag{2.5.2}$$

With the preceding discussion, we can now present the variational formula for the Yang-Mills-Higgs connection.

Lemma 2.6. *Suppose (A, ϕ) is Yang-Mills-Higgs connection, then*

$$\begin{aligned} 0 &= \int_M \left((|F_A|^2 + |D^{as} \phi|^2 + V(\phi)) \operatorname{div} X - 4 \sum_{i,j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \right. \\ &\quad \left. + 2 \sum_{i=1}^n \left(\langle D_{\nabla_X e_i}^{as} \phi, D_{e_i}^{as} \phi \rangle - \langle D_{\nabla_{e_i} X}^{as} \phi, D_{e_i}^{as} \phi \rangle \right) \right) dV_g. \end{aligned} \tag{2.5.3}$$

Proof. We start from the expression for the Yang-Mills-Higgs functional for the transformed fields (A^t, ϕ^t) given in equation (2.5.2) of the provided text:

$$\begin{aligned} \mathcal{YM}\mathcal{H}(A^t, \phi^t) &= \frac{1}{4\pi^2} \int_M \left(\underbrace{\sum_{i,j} |F_A(d\psi_t(e_i(\psi_t^{-1}(x))), d\psi_t(e_j(\psi_t^{-1}(x))))|^2}_{K_t(x)} \right. \\ &\quad \left. + \underbrace{\sum_i |D_{d\psi_t(e_i(\psi_t^{-1}(x)))}^{as} \phi(x)|^2}_{H_t(x)} + \underbrace{V(\phi(x))}_{P_t(x)} \right) \underbrace{|\det(d\psi_t^{-1}(x))|}_{J_t(x)} dV_g(x). \end{aligned}$$

Let $L_t(x) = K_t(x) + H_t(x) + P_t(x)$, then the functional is

$$\mathcal{YM}\mathcal{H}(A^t, \phi^t) = \frac{1}{4\pi^2} \int_M L_t(x) J_t(x) dV_g(x).$$

We differentiate the integrand with respect to t at $t = 0$ using the product rule:

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} (L_t(x) J_t(x)) &= \left(\left. \frac{dK_t}{dt} \right|_{t=0} + \left. \frac{dH_t}{dt} \right|_{t=0} + \left. \frac{dP_t}{dt} \right|_{t=0} \right) J_0(x) + L_0(x) \left(\left. \frac{dJ}{dt} \right|_{t=0} \right) \\ &= \left(\left. \frac{dK_t}{dt} \right|_{t=0} + \left. \frac{dH_t}{dt} \right|_{t=0} + \left. \frac{dP_t}{dt} \right|_{t=0} \right) \\ &\quad + (|F_A|^2(x) + |D^{as}\phi|^2(x) + V(\phi(x))) \left(\left. \frac{dJ}{dt} \right|_{t=0} \right). \end{aligned} \quad (2.5.4)$$

Let $X = \left. \frac{d\psi_t}{dt} \right|_{t=0}$ be the vector field generating the flow. Calculate Derivatives at $t = 0$ respectively,

- **Jacobian Term** $\left. \frac{dJ_t}{dt} \right|_{t=0}$:

$$\begin{aligned} \left. \frac{dJ_t}{dt} \right|_{t=0} &= \left. \frac{d}{dt} \right|_{t=0} \det(d\phi_t^{-1}) \\ &= \det(d\phi_0^{-1}) \text{Tr} \left((d\phi_0^{-1})^{-1} \left. \frac{d}{dt} \right|_{t=0} (d\phi_t^{-1}) \right) \\ &= \text{Tr} \left(d \left(\left. \frac{d}{dt} \right|_{t=0} \phi_t^{-1} \right) \right) \\ &= \text{Tr}(d(-X)) \\ &= -\text{div} X \end{aligned}$$

- **Curvature Term** $\left. \frac{dK_t}{dt} \right|_{t=0}$:

Denote $u_i(t, x)$ by $d\psi_t(e_i(\psi_t^{-1}(x)))$, then $e_i = u_i|_{t=0}$. We use an identity $v_i := \left. \frac{du_i}{dt} \right|_{t=0} = -[X, e_i]$ in the following computation, which can be verified directly via the definition of Lie derivative.

$$\begin{aligned} \left. \frac{dK_t}{dt} \right|_{t=0} &= \sum_{i,j} 2 \left\langle \left. \frac{d}{dt} \right|_{t=0} F_A(u_i(t, x), u_j(t, x)), F_A(u_i(0, x), u_j(0, x)) \right\rangle \\ &= \sum_{i,j} 2 \left\langle F_A \left(\left. \frac{du_i}{dt} \right|_{t=0}, u_j(0) \right) + F_A \left(u_i(0), \left. \frac{du_j}{dt} \right|_{t=0} \right), F_A(u_i(0, x), u_j(0, x)) \right\rangle \\ &= \sum_{i,j} 2 \langle F_A(v_i, e_j) + F_A(e_i, v_j), F_A(e_i, e_j) \rangle \\ &= 4 \sum_{i,j} \langle F_A(v_i, e_j), F_A(e_i, e_j) \rangle \\ &= -4 \sum_{i,j} \langle F_A([X, e_i], e_j), F_A(e_i, e_j) \rangle. \end{aligned}$$

• **Higgs Kinetic Term** $\frac{dH_t}{dt}|_{t=0}$:

We use linearity of D^{as} in the subscript vector field and identity $\frac{du_i}{dt}|_{t=0} = -[X, e_i]$ to compute

$$\begin{aligned} \frac{dH_t}{dt}\Big|_{t=0} &= \sum_i 2 \left\langle \frac{d}{dt}\Big|_{t=0} (D_{u_i(t,x)}^{as} \phi(x)), D_{u_i(0,x)}^{as} \phi(x) \right\rangle \\ &= \sum_i 2 \left\langle D_{\frac{du_i}{dt}|_{t=0}}^{as} \phi(x), D_{e_i(x)}^{as} \phi(x) \right\rangle \\ &= -2 \sum_i \langle D_{[X, e_i]}^{as} \phi, D_{e_i}^{as} \phi \rangle \end{aligned}$$

• **Potential Term** $\frac{dP_t}{dt}|_{t=0}$:

Since this expression $P_t(x) = V(\phi(x))$ does not explicitly depend on t , we have $\frac{dP_t}{dt}|_{t=0} = 0$.

Substitute the derivatives back into the formula (2.5.4), we obtain

$$\begin{aligned} \frac{d}{dt}\Big|_{t=0} (L_t(x) J_t(x)) &= -(|F_A|^2 + |D^{as} \phi|^2 + V(\phi)) \operatorname{div} X \\ &\quad - 4 \sum_{i,j} \langle F_A([X, e_i], e_j), F_A(e_i, e_j) \rangle - 2 \sum_i \langle D_{[X, e_i]}^{as} \phi, D_{e_i}^{as} \phi \rangle. \end{aligned}$$

Integrate over M and multiplying by the constant factor yields the first variation of the functional:

$$\begin{aligned} \frac{d}{dt}\Big|_{t=0} \mathcal{YM}\mathcal{H}(A^t, \phi^t) &= -\frac{1}{4\pi^2} \int_M \left((|F_A|^2 + |D^{as} \phi|^2 + V(\phi)) \operatorname{div} X \right. \\ &\quad \left. + 4 \sum_{i,j} \langle F_A([X, e_i], e_j), F_A(e_i, e_j) \rangle + 2 \sum_i \langle D_{[X, e_i]}^{as} \phi, D_{e_i}^{as} \phi \rangle \right) dV_g. \end{aligned} \tag{2.5.5}$$

Since $[X, e_i] = \nabla_X e_i - \nabla_{e_i} X$, where ∇ denotes the Levi-Civita connection of g , we obtain

$$\begin{aligned} \sum_{i,j=1}^n \langle F_A([X, e_i], e_j), F_A(e_i, e_j) \rangle &= - \sum_{i,j=1}^n (\langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \\ &\quad - \langle F_A(\nabla_X e_i, e_j), F_A(e_i, e_j) \rangle) \\ &= - \sum_{i,j=1}^n (\langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \\ &\quad - g(\nabla_X e_i, e_k) \langle F_A(e_k, e_j), F_A(e_i, e_j) \rangle) \\ &= - \sum_{i,j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle, \end{aligned}$$

where in the last step, we used $g(\nabla_X e_i, e_k) = -g(e_i, \nabla_X e_k) = -g(\nabla_X e_k, e_i)$, and

$$\sum_i \langle D_{[X, e_i]}^{as} \phi, D_{e_i}^{as} \phi \rangle = \sum_i \left(\langle D_{\nabla_X e_i}^{as} \phi, D_{e_i}^{as} \phi \rangle - \langle D_{\nabla_{e_i} X}^{as} \phi, D_{e_i}^{as} \phi \rangle \right).$$

Combine this computations with (2.5.5) and the fact that (A, ϕ) is Yang-Mills connection, we obtain

$$\begin{aligned} 0 = \int_M & \left((|F_A|^2 + |D^{as} \phi|^2 + V(\phi)) \operatorname{div} X - 4 \sum_{i,j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \right. \\ & \left. + 2 \sum_{i=1}^n \left(\langle D_{\nabla_X e_i}^{as} \phi, D_{e_i}^{as} \phi \rangle - \langle D_{\nabla_{e_i} X}^{as} \phi, D_{e_i}^{as} \phi \rangle \right) \right) dV_g. \end{aligned}$$

□

This formula (2.5.3) is the general formula for the variation of the Yang-Mills-Higgs functional. Each term corresponds to a different contribution from the curvature, the Higgs field, and the potential, and the divergence term accounts for the flow generated by the vector field X .

Part II: a monotonicity formula for YMH connection

Now we derive a monotonicity formula via above variational lemma. More specifically, we will analyze the monotonicity properties of a weighted energy functional

$$\mathcal{E}_{YMH}(\tau) = e^{a\tau^2} \int_{B_\tau(p)} (|F_A|^2 + |D^{as} \phi|^2 + V(\phi)) dV_g.$$

Fix any $p \in M$, let r_p be a positive number with properties: there are normal coordinates $x_1, \dots, x_n$ in the geodesic ball $B_{r_p}(p)$ of (M, g) , such that $p = (0, \dots, 0)$ and for some constant $C(p)$,

$$\begin{aligned} |g_{ij} - \delta_{ij}| &\leq C(p)(|x_1|^2 + \dots + |x_n|^2), \\ |dg_{ij}| &\leq C(p)\sqrt{|x_1|^2 + \dots + |x_n|^2}. \end{aligned}$$

The constants r_p and $C(p)$ can be chosen depending only on the injective radius at p and the curvature of g . If $M = \mathbb{R}^n$ and g is flat, we can take $r_p = \infty$ and $C(p) = 0$.

Let $r(x)$ be the distance function from p ; i.e. $r(x) = \sqrt{x_1^2 + \dots + x_n^2}$, α be a positive function on the unit sphere S^{n-1} , define a vector field

$$X(x) = \xi(r) \alpha \left(\frac{x}{r} \right) r \frac{\partial}{\partial r} = \xi(r) \alpha \left(\frac{x}{r} \right) \left(\sum_i x_i \frac{\partial}{\partial x_i} \right),$$

where ξ is some smooth function with compact support in $B_{r_p}(p)$.

Let $\{e_1, \dots, e_n\}$ be any orthonormal basis near p such that $e_1 = \frac{\partial}{\partial r}$. Since $x_1, \dots, x_n$ are normal coordinates, we have $\nabla_{\frac{\partial}{\partial r}} \frac{\partial}{\partial r} = 0$, thus

$$\begin{aligned} \nabla_{\frac{\partial}{\partial r}} X &= \nabla_{\frac{\partial}{\partial r}} \left(\xi(r) \alpha(\theta) r \frac{\partial}{\partial r} \right) \\ &= \frac{\partial}{\partial r} (\xi(r) \alpha(\theta) r) \frac{\partial}{\partial r} \\ &= (\xi' r + \xi) \alpha(\theta) \frac{\partial}{\partial r}, \end{aligned} \quad (2.5.6)$$

where $\theta = \frac{x}{r}$. Moreover, for $i \geq 2$,

$$\begin{aligned} \nabla_{e_i} X &= \nabla_{e_i} \left(\xi(r) \alpha(\theta) r \frac{\partial}{\partial r} \right) \\ &= e_i (\xi \alpha(\theta) r) \frac{\partial}{\partial r} + \xi \alpha(\theta) r \nabla_{e_i} \frac{\partial}{\partial r} \\ &= \xi r e_i (\alpha(\theta)) \frac{\partial}{\partial r} + \xi \alpha(\theta) \sum_{j=1}^n b_{ij} e_j. \end{aligned} \quad (2.5.7)$$

One can readily check that for the coefficients b_{ij} , we have asymptotic estimates (we always denote by $O(1)$ a quantity bounded by a constant depending only on n .)

$$b_{ij} = \delta_{ij} + O(1)C(p)r^2.$$

Actually, let $c_{ij} = g(\nabla_{e_i} \frac{\partial}{\partial r}, e_j)$ be the components of $\nabla_{e_i} \frac{\partial}{\partial r}$ in the orthonormal basis $\{e_j\}$, then $b_{ij} = r c_{ij}$. Clearly, $c_{i1} = c_{1j} = 0$ and $c_{ij} = II(e_i, e_j)$ are the components of the second fundamental form of the geodesic sphere S_r . Moreover, $c_{ij} = \frac{1}{r} \delta_{ij}$ in flat space and $c_{ij} \approx \frac{1}{r} \delta_{ij} - \frac{r}{3} R_{1i1j} + O(r^2)$ in curved space, thus for b_{ij} , we have

1. $b_{i1} = b_{1j} = 0$.
2. For $i, j \geq 2$, $b_{ij} = \delta_{ij}$ in flat space and $b_{ij} \approx \delta_{ij} - \frac{r^2}{3} R_{1i1j} + O(r^3)$ in curved space.

We now utilize equations (2.5.6) and (2.5.7) in conjunction with the first variational formula (2.5.3) to derive a new equation.

Rewrite equation (2.5.3) as:

$$\begin{aligned} \int_M (|F_A|^2 + |D^{as} \phi|^2 + V(\phi)) \operatorname{div} X \, dV_g &= \int_M \left(4 \sum_{i,j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle \right. \\ &\quad \left. - 2 \sum_{i=1}^n \left(\langle D_{\nabla_X e_i}^{as} \phi, D_{e_i}^{as} \phi \rangle - \langle D_{\nabla_{e_i} X}^{as} \phi, D_{e_i}^{as} \phi \rangle \right) \right) dV_g. \end{aligned} \quad (2.5.8)$$

Observe that

$$\begin{aligned}
 \operatorname{div} X &= \sum_{i=1}^n g(\nabla_{e_i} X, e_i) \\
 &= (\xi' r + \xi) \alpha + \xi \alpha \sum_{i=2}^n b_{ii} \\
 &= (\xi' r + n\xi) \alpha + O(1)C(p)r^2 \xi \alpha.
 \end{aligned} \tag{2.5.9}$$

Regarding $S = 4 \sum_{i,j} \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle$, we split this into two parts:

- Case $i = 1$:

$$S_1 = 4 \sum_{j=1}^n \langle F_A(\nabla_{e_1} X, e_j), F_A(e_1, e_j) \rangle,$$

- Case $i \geq 2$:

$$S_2 = 4 \sum_{i=2}^n \sum_{j=1}^n \langle F_A(\nabla_{e_i} X, e_j), F_A(e_i, e_j) \rangle.$$

The computation of S_1 is straightforward,

$$\begin{aligned}
 S_1 &= 4 \sum_{j=1}^n \langle F_A(\nabla_{e_1} X, e_j), F_A(e_1, e_j) \rangle \\
 &= 4(\xi' r + \xi) \alpha \sum_{j=2}^n |F_A(e_1, e_j)|^2 \\
 &= 4(\xi' r + \xi) \alpha |\iota_{e_1} F_A|^2.
 \end{aligned} \tag{2.5.10}$$

Split S_2 into (notice that $e_1(\alpha) = 0$)

$$\begin{aligned}
 S_{2a} &= 4\xi r \sum_{i=2}^n \sum_{j=1}^n e_i(\alpha) \langle F_A(e_1, e_j), F_A(e_i, e_j) \rangle \\
 &= 4\xi r \sum_{j=1}^n \left(\langle F_A(e_1, e_j), \sum_{i=1}^n e_i(\alpha) F_A(e_i, e_j) \rangle \right) \\
 &= 4\xi r \langle \iota_{e_1} F_A, \iota_{\nabla \alpha} F_A \rangle
 \end{aligned}$$

and

$$\begin{aligned}
 S_{2b} &= 4\xi \alpha \sum_{i=2}^n \sum_{j=1}^n \sum_{l=1}^n b_{il} \langle F_A(e_l, e_j), F_A(e_i, e_j) \rangle \\
 &= 4\xi \alpha \sum_{i=2}^n \sum_{j=1}^n \sum_{l=1}^n (\delta_{il} + O(1)C(p)r^2) \langle F_A(e_l, e_j), F_A(e_i, e_j) \rangle \\
 &:= S_{2b,\delta} + S_{2b,O(r^2)},
 \end{aligned}$$

where the leading term $S_{2b,\delta}$ and the residual term $S_{2b,O(r^2)}$ are

$$S_{2b,\delta} = 4\xi\alpha \sum_{i=2}^n \sum_{j=1}^n \langle F_A(e_i, e_j), F_A(e_i, e_j) \rangle;$$

$$S_{2b,O(r^2)} = 4\xi\alpha \sum_{i=2}^n \sum_{j=1}^n \sum_{l=1}^n O(1)C(p)r^2 \langle F_A(e_l, e_j), F_A(e_i, e_j) \rangle.$$

For the leading term $S_{2b,\delta}$,

$$\begin{aligned} 4\xi\alpha \sum_{i=2}^n \sum_{j=1}^n \langle F_A(e_i, e_j), F_A(e_i, e_j) \rangle &= 4\xi\alpha \sum_{i=2}^n |F_A(e_i, \cdot)|^2 \\ &= 4\xi\alpha \left(\sum_{i=1}^n |F_A(e_i, \cdot)|^2 - |F_A(e_1, \cdot)|^2 \right) \\ &= 4\xi\alpha (|F_A|^2 - |\iota_{e_1} F_A|^2) \end{aligned}$$

Hence

$$S_2 = 4\xi r \langle \iota_{e_1} F_A, \iota_{\nabla_X} F_A \rangle + 4\xi\alpha (|F_A|^2 - |\iota_{e_1} F_A|^2) + \xi\alpha O(1)C(p)r^2 |F_A|^2. \quad (2.5.11)$$

It remains to compute the asymptotic expansion of the term

$$\sum_{i=1}^n \left(\langle D_{\nabla_X e_i}^{as} \phi, D_{e_i}^{as} \phi \rangle - \langle D_{\nabla_{e_i} X}^{as} \phi, D_{e_i}^{as} \phi \rangle \right)$$

in equation (2.5.8). We still divide the sum into the cases $i = 1$ and $i \geq 2$.

- Case $i = 1$:

We use the fact from (2.5.6) and $\nabla_X e_1 = 0$ to compute

$$\langle D_{\nabla_X e_1}^{as} \phi, D_{e_1}^{as} \phi \rangle - \langle D_{\nabla_{e_1} X}^{as} \phi, D_{e_1}^{as} \phi \rangle = -(\xi' r + \xi)\alpha |D_{e_1}^{as} \phi|^2.$$

- Case $i \geq 2$:

We first compute $\nabla_X e_i$ via (2.5.7) and the structure functions C_{ij}^k (coefficients of the Lie bracket relative to the frame $\{e_i\}$) of the base manifold,

$$\begin{aligned} \nabla_X e_i &= \xi\alpha r \nabla_{e_1} e_i \\ &= \xi\alpha r (\nabla_{e_i} e_1 + [e_1, e_i]) \\ &= \xi\alpha e_i + \xi\alpha C(p)r^2 \sum_j O(1)e_j + \xi\alpha r \sum_j C_{1i}^j e_j. \end{aligned}$$

Plug this and (2.5.7) into the original term, we obtain (recall near the point p , $C_{ij}^k = O(r)$.)

$$\begin{aligned}
 & \sum_{i=2}^n \left(\langle D_{\nabla_X e_i}^{as} \phi, D_{e_i}^{as} \phi \rangle - \langle D_{\nabla_{e_i} X}^{as} \phi, D_{e_i}^{as} \phi \rangle \right) \\
 &= \sum_{i=2}^n \left\langle D_{\xi \alpha e_i + \xi \alpha C(p) r^2 \sum_j O(1) e_j + \xi \alpha r \sum_j C_{1i}^j e_j}^{as} \phi, D_{e_i}^{as} \phi \right\rangle \\
 & \quad - \sum_{i=2}^n \left\langle D_{\xi r e_i(\alpha) e_1 + \xi \alpha e_i + \xi \alpha C(p) r^2 \sum_j O(1) e_j}^{as} \phi, D_{e_i}^{as} \phi \right\rangle \\
 &= \xi \alpha \sum_{i=2}^n |D_{e_i}^{as} \phi|^2 + \xi \alpha \sum_{i=2}^n \sum_{j=1}^n (C(p) r^2 O(1) + r C_{1i}^j) \langle D_{e_j}^{as} \phi, D_{e_i}^{as} \phi \rangle \\
 & \quad - \xi r \sum_{i=2}^n e_i(\alpha) \langle D_{e_1}^{as} \phi, D_{e_i}^{as} \phi \rangle - \xi \alpha \sum_{i=2}^n |D_{e_i}^{as} \phi|^2 - \xi \alpha C(p) r^2 \sum_{i=2}^n \sum_{j=1}^n O(1) \langle D_{e_j}^{as} \phi, D_{e_i}^{as} \phi \rangle \\
 &= -\xi r \langle D_{e_1}^{as} \phi, D_{\nabla \alpha}^{as} \phi \rangle + \xi \alpha C(p) r^2 \sum_{i=2}^n \sum_{j=1}^n O(1) \langle D_{e_j}^{as} \phi, D_{e_i}^{as} \phi \rangle.
 \end{aligned}$$

Hence, we conclude that

$$\begin{aligned}
 \sum_{i=1}^n \left(\langle D_{\nabla_X e_i}^{as} \phi, D_{e_i}^{as} \phi \rangle - \langle D_{\nabla_{e_i} X}^{as} \phi, D_{e_i}^{as} \phi \rangle \right) &= -(\xi' r + \xi) \alpha |D_{e_1}^{as} \phi|^2 - \xi r \langle D_{e_1}^{as} \phi, D_{\nabla \alpha}^{as} \phi \rangle \\
 & \quad + \xi \alpha C(p) r^2 \sum_{i=2}^n \sum_{j=1}^n O(1) \langle D_{e_j}^{as} \phi, D_{e_i}^{as} \phi \rangle.
 \end{aligned} \tag{2.5.12}$$

Plug (2.5.9)-(2.5.12) into (2.5.8), we obtain

$$\begin{aligned}
 & \int_M (|F_A|^2 + |D^{as} \phi|^2 + V(\phi)) ((\xi' r + n\xi) \alpha + O(1) C(p) r^2 \xi \alpha) dV_g \\
 &= \int_M \left(4(\xi' r + \xi) \alpha |\iota_{e_1} F_A|^2 + 4\xi r \langle \iota_{e_1} F_A, \iota_{\nabla \alpha} F_A \rangle + 4\xi \alpha (|F_A|^2 - |\iota_{e_1} F_A|^2) + \xi \alpha O(1) C(p) r^2 |F_A|^2 \right. \\
 & \quad + 2(\xi' r + \xi) \alpha |D_{e_1}^{as} \phi|^2 + 2\xi r \langle D_{e_1}^{as} \phi, D_{\nabla \alpha}^{as} \phi \rangle \\
 & \quad \left. - 2\xi \alpha C(p) r^2 \sum_{i=2}^n \sum_{j=1}^n O(1) \langle D_{e_j}^{as} \phi, D_{e_i}^{as} \phi \rangle \right) dV_g.
 \end{aligned} \tag{2.5.8}$$

Simplify this identity yields

$$\begin{aligned}
 & \int_M \left(|F_A|^2 (\xi' r + (n-4)\xi + O(1)C(p)r^2\xi)\alpha \right. \\
 & \quad + |D^{as}\phi|^2 ((\xi' r + n\xi)\alpha + O(1)C(p)r^2\xi\alpha) \\
 & \quad \left. + V(\phi) ((\xi' r + n\xi)\alpha + O(1)C(p)r^2\xi\alpha) \right) dV_g \\
 & = 4 \int_M \left(\xi' r \alpha \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 + \xi r \langle \iota_{\frac{\partial}{\partial r}} F_A, \iota_{\nabla \alpha} F_A \rangle \right. \\
 & \quad \left. + \frac{1}{2} (\xi' r + \xi) \alpha |D_{e_1}^{as}\phi|^2 + \frac{1}{2} \xi r \langle D_{e_1}^{as}\phi, D_{\nabla \alpha}^{as}\phi \rangle \right) dV_g. \tag{2.5.9}
 \end{aligned}$$

We turn to specific selections for the functions α and ξ to analyze identity (2.5.9). First put $\alpha = 1$. For any τ small enough, put $\xi(r) = \xi_\tau(r) = \eta(r/\tau)$, where η is smooth and satisfies: $\eta(r) = 1$ for $r \in [0, 1]$, $\eta(r) = 0$ for $r \in [1 + \varepsilon, \infty)$, $\varepsilon > 0$ and $\eta'(r) \leq 0$. Then $\tau \frac{\partial}{\partial \tau}(\xi_\tau(r)) = -r\xi'_\tau(r)$, plug these into (2.5.9), we obtain (for $\tau \geq r$)

$$\begin{aligned}
 & \tau \frac{d}{d\tau} \left(\int_M \xi_\tau |F_A|^2 dV_g \right) + ((4-n) + O(1)C(p)\tau^2) \int_M \xi_\tau |F_A|^2 dV_g \\
 & \tau \frac{d}{d\tau} \left(\int_M \xi_\tau |D^{as}\phi|^2 dV_g \right) + (-n + O(1)C(p)\tau^2) \int_M \xi_\tau |D^{as}\phi|^2 dV_g \\
 & = 4\tau \frac{d}{d\tau} \int_M \xi_\tau \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 dV_g + 2\tau \frac{d}{d\tau} \int_M \xi_\tau |D_{e_1}^{as}\phi|^2 dV_g - 2 \int_M \xi_\tau |D_{e_1}^{as}\phi|^2 dV_g \\
 & \quad - \tau \frac{d}{d\tau} \int_M \xi_\tau V(\phi) dV_g + n \int_M \xi_\tau V(\phi) dV_g + O(1)C(p)\tau^2 \int_M V(\phi) \xi_\tau dV_g. \tag{2.5.10}
 \end{aligned}$$

We adopt the following notation for simplicity:

$$\begin{aligned}
 I(\tau) &= \int_M \xi_\tau |F_A|^2 dV_g; & J(\tau) &= \int_M \xi_\tau \left| \iota_{\frac{\partial}{\partial r}} F_A \right|^2 dV_g; \\
 H(\tau) &= \int_M \xi_\tau |D^{as}\phi|^2 dV_g; & N(\tau) &= \int_M \xi_\tau |D_{e_1}^{as}\phi|^2 dV_g; \\
 P(\tau) &= \int_M \xi_\tau V(\phi) dV_g; & \epsilon(\tau) &= O(1)C(p)\tau^2.
 \end{aligned}$$

then the starting equation (2.5.10) reads

$$\begin{aligned}
 & \tau \frac{d}{d\tau} I(\tau) + (4-n)I(\tau) + \epsilon(\tau)I(\tau) + \tau \frac{d}{d\tau} H(\tau) - nH(\tau) + \epsilon(\tau)H(\tau) \\
 & = 4\tau \frac{d}{d\tau} J(\tau) + 2\tau \frac{d}{d\tau} N(\tau) - 2N(\tau) - \tau \frac{d}{d\tau} P(\tau) + nP(\tau) + \epsilon(\tau)P(\tau). \tag{2.5.11}
 \end{aligned}$$

Let $L_1(\tau) = \tau^{4-n} e^{a\tau^2} I(\tau)$, $L_2(\tau) = \tau^{-n} e^{a\tau^2} H(\tau)$ and $L_3(\tau) = e^{a\tau^2} P(\tau)$, the parameter a

will be specified later. We compute $\frac{dL_1}{d\tau}$, $\frac{dL_2}{d\tau}$ and $\frac{dL_3}{d\tau}$ using the product rule:

$$\begin{aligned}\frac{dL_1}{d\tau} &= ((4-n)\tau^{3-n})e^{a\tau^2}I(\tau) + \tau^{4-n}\left(e^{a\tau^2}(2a\tau)\right)I(\tau) + \tau^{4-n}e^{a\tau^2}\frac{dI}{d\tau} \\ &= \tau^{4-n}e^{a\tau^2}\left(\frac{1}{\tau}\left((4-n)I(\tau) + (2a\tau^2)I(\tau) + \tau\frac{dI}{d\tau}\right)\right). \\ \frac{dL_2}{d\tau} &= (-n\tau^{-n-1})e^{a\tau^2}H(\tau) + \tau^{-n}\left(e^{a\tau^2}(2a\tau)\right)H(\tau) + \tau^{-n}e^{a\tau^2}\frac{dH}{d\tau} \\ &= \tau^{-n}e^{a\tau^2}\left(\frac{1}{\tau}\left(-nH(\tau) + (2a\tau^2)H(\tau) + \tau\frac{dH}{d\tau}\right)\right). \\ \frac{dL_3}{d\tau} &= \left(e^{a\tau^2}(2a\tau)\right)P(\tau) + e^{a\tau^2}\frac{dP}{d\tau} \\ &= e^{a\tau^2}\left(2a\tau P(\tau) + \frac{dP}{d\tau}\right).\end{aligned}$$

From (2.5.11), we solve for $\tau\frac{dI}{d\tau} + \tau\frac{dH}{d\tau} + \tau\frac{d}{d\tau}P(\tau)$,

$$\begin{aligned}\tau\frac{d}{d\tau}I(\tau) + \tau\frac{d}{d\tau}H(\tau) + \tau\frac{d}{d\tau}P(\tau) &= 4\tau\frac{d}{d\tau}J(\tau) + 2\tau\frac{d}{d\tau}N(\tau) - 2N(\tau) + nP(\tau) + \epsilon(\tau)P(\tau) \\ &\quad + (n-4)I(\tau) - \epsilon(\tau)I(\tau) + nH(\tau) - \epsilon(\tau)H(\tau),\end{aligned}$$

substitute this into the expression for $\tau^{n-4}\frac{dL_1}{d\tau} + \tau^n\frac{dL_2}{d\tau} + \frac{dL_3}{d\tau}$ and compute

$$\begin{aligned}\tau^{n-4}\frac{dL_1}{d\tau} + \tau^n\frac{dL_2}{d\tau} + \frac{dL_3}{d\tau} &= e^{a\tau^2}\left((2a - O(1)C(p))\tau(I + H) + (2a + O(1)C(p))\tau P\right. \\ &\quad \left.+ 4\frac{dJ}{d\tau} + 2\frac{dN}{d\tau} - 2\frac{N(\tau)}{\tau} + n\frac{P(\tau)}{\tau}\right).\end{aligned}\tag{2.5.12}$$

Choose the nonnegative constant $a \geq \max\{1/2O(1)C(p), -1/2O(1)C(p)\}$, we deduce from the above

$$\begin{aligned}\tau^{n-4}\frac{d}{d\tau}\left(\tau^{4-n}e^{a\tau^2}\int_M \xi_\tau |F_A|^2 dV_g\right) &+ \tau^n\frac{d}{d\tau}\left(\tau^{-n}e^{a\tau^2}\int_M \xi_\tau |D^{as}\phi|^2 dV_g\right) + \frac{d}{d\tau}\left(e^{a\tau^2}\int_M \xi_\tau V(\phi)dV_g\right) \\ &= e^{a\tau^2}\left((2a - O(1)C(p))\tau(I + H) + (2a + O(1)C(p))\tau P\right. \\ &\quad \left.+ 4\frac{d}{d\tau}\int_M \xi_\tau \left|\iota_{\frac{\partial}{\partial\tau}}F_A\right|^2 dV_g + 2\frac{d}{d\tau}\int_M \xi_\tau |D_{e_1}^{as}\phi|^2 dV_g - 2\frac{N(\tau)}{\tau} + n\frac{P(\tau)}{\tau}\right).\end{aligned}\tag{2.5.13}$$

The coefficient of $\tau(I+H)$ is $(2a - O(1)C(p)) \geq 0$, the coefficient of τP is $(2a + O(1)C(p)) \geq 0$, since $I, H, P \geq 0$, the first two terms in RHS of (2.5.13) is non-negative, thus:

$$\tau^{n-4}\frac{dL_1}{d\tau} + \tau^n\frac{dL_2}{d\tau} + \frac{dL_3}{d\tau} \geq e^{a\tau^2}\left(4\frac{dJ}{d\tau} + 2\frac{dN}{d\tau} - 2\frac{N(\tau)}{\tau} + n\frac{P(\tau)}{\tau}\right).$$

Integrate the inequality from σ to ρ , where $0 < \sigma < \rho$, we obtain (recall dimension $n \geq 4$.)

$$\begin{aligned} \int_{\sigma}^{\rho} \frac{d}{d\tau} (\tau^{n-4} L_1 + \tau^n L_2 + L_3) d\tau &= \int_{\sigma}^{\rho} \left(\tau^{n-4} \frac{dL_1}{d\tau} + \tau^n \frac{dL_2}{d\tau} + \frac{dL_3}{d\tau} \right) + (n-4)\tau^{n-5} L_1 + n\tau^{n-1} L_2 d\tau \\ &\geq \int_{\sigma}^{\rho} e^{a\tau^2} \left(4 \frac{dJ}{d\tau} + 2 \frac{dN}{d\tau} - 2 \frac{N(\tau)}{\tau} + n \frac{P(\tau)}{\tau} \right) d\tau. \end{aligned} \quad (2.5.14)$$

Taking $\varepsilon \rightarrow 0$ in inequality (2.5.14), we study the limiting behavior and determine the resulting conclusions. As $\varepsilon \rightarrow 0$ in the cutoff function $\xi_{\tau} = \eta(r/\tau)$ implies $\xi_{\tau} \rightarrow \chi_{B_{\tau}(p)}$, the integrals become integrals over the ball $B_{\tau}(p)$, i.e.

- $I(\tau) = \int_M \xi_{\tau} |F_A|^2 dV_g \longrightarrow \int_{B_{\tau}(p)} |F_A|^2 dV_g,$
- $H(\tau) = \int_M \xi_{\tau} |D^{as}\phi|^2 dV_g \longrightarrow \int_{B_{\tau}(p)} |D^{as}\phi|^2 dV_g,$
- $P(\tau) = \int_M \xi_{\tau} V(\phi) dV_g \longrightarrow \int_{B_{\tau}(p)} V(\phi) dV_g,$
- $N(\tau) = \int_M \xi_{\tau} |D_{e_1}^{as}\phi|^2 dV_g \longrightarrow \int_{B_{\tau}(p)} |D_{e_1}^{as}\phi|^2 dV_g,$
- $P(\tau) = \int_M \xi_{\tau} V(\phi) dV_g \longrightarrow \int_{B_{\tau}(p)} V(\phi) dV_g.$

For the J term, we claim a fact involving $dJ/d\tau$:

$$\lim_{\varepsilon \rightarrow 0} \frac{dJ}{d\tau} = \int_{\partial B_{\tau}(p)} |\iota_{e_1} F_A|^2 dS_{\tau}. \quad (2.5.15)$$

proof of the claim. Let's calculate $\frac{dJ}{d\tau}$ precisely using the definition $J(\tau) = \int_M \xi_{\tau}(r) f(x) dV_g$, where $f(x) = |\iota_{e_1} F_A|^2(x)$. Denote $H(r) = \int_{\partial B_r(p)} f(y) dS_r(y)$ (here $H(r)$ does not depend on τ), then

$$J(\tau) = \int_0^{\infty} \eta(r/\tau) H(r) dr$$

and

$$\frac{dJ}{d\tau} = \int_0^{\infty} \frac{\partial}{\partial \tau} (\eta(r/\tau)) H(r) dr = -\frac{1}{\tau^2} \int_0^{\infty} r \eta'(r/\tau) H(r) dr$$

Now, consider the limit $\varepsilon \rightarrow 0$. Obviously, the smooth function $\eta(s)$ approaches the Heaviside step function $H(1-s)$ (1 for $s \leq 1$, 0 for $s > 1$) as $\varepsilon \rightarrow 0$. Correspondingly, its derivative $\eta'(s)$ approaches the negative Dirac delta function centered at $s = 1$: $\eta'(s) \rightarrow -\delta(s-1)$. In this limit, the expression for $\frac{dJ}{d\tau}$ becomes:

$$\lim_{\varepsilon \rightarrow 0} \frac{dJ}{d\tau} = -\frac{1}{\tau^2} \int_0^{\infty} r \left(-\delta\left(\frac{r}{\tau} - 1\right) \right) H(r) dr.$$

Using the property $\delta(ax-b) = \frac{1}{|a|} \delta(x-b/a)$, we have $\delta(r/\tau - 1) = \tau \delta(r - \tau)$, thus

$$\lim_{\varepsilon \rightarrow 0} \frac{dJ}{d\tau} = \frac{1}{\tau} \int_0^{\infty} r \delta(r - \tau) H(r) dr = \frac{1}{\tau} (\tau H(\tau)) = H(\tau) = \int_{\partial B_{\tau}(p)} |\iota_{e_1} F_A|^2 dS_{\tau}.$$

That is (2.5.15). Thanks to (2.5.15), we have

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \int_{\sigma}^{\rho} 4e^{a\tau^2} \frac{dJ}{d\tau} d\tau &= \int_{\sigma}^{\rho} 4e^{a\tau^2} \left(\int_{\partial B_{\tau}(p)} |\iota_{e_1} F_A|^2 dS_{\tau} \right) d\tau \\ &= \int_{B_{\rho}(p) \setminus B_{\sigma}(p)} 4e^{a\tau^2} |\iota_{e_1} F_A|^2 dV_g. \end{aligned}$$

For the N term, Similarly, we have

$$\lim_{\varepsilon \rightarrow 0} \frac{dN}{d\tau} = \int_{\partial B_{\tau}(p)} |D_{e_1}^{as} \phi|^2 dS_{\tau} \quad (2.5.16)$$

and

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \int_{\sigma}^{\rho} 2e^{a\tau^2} \frac{dN}{d\tau} d\tau &= \int_{\sigma}^{\rho} 2e^{a\tau^2} \left(\int_{\partial B_{\tau}(p)} |D_{e_1}^{as} \phi|^2 dS_{\tau} \right) d\tau \\ &= \int_{B_{\rho}(p) \setminus B_{\sigma}(p)} 2e^{a\tau^2} |D_{e_1}^{as} \phi|^2 dV_g. \end{aligned}$$

Substitute all the limits we have concluded above into the inequality (2.5.14), we conclude the following monotonicity formula for Yang-Mills-Higgs connection

Theorem 2.7. *Let r_p , a be as above. Then for any $0 < \sigma < \rho < r_p$, we have*

$$\begin{aligned} &e^{a\rho^2} \left(\int_{B_{\rho}(p)} |F_A|^2 dV_g + \int_{B_{\rho}(p)} |D^{as} \phi|^2 dV_g + \int_{B_{\rho}(p)} V(\phi) dV_g \right) \\ &- e^{a\sigma^2} \left(\int_{B_{\sigma}(p)} |F_A|^2 dV_g + \int_{B_{\sigma}(p)} |D^{as} \phi|^2 dV_g + \int_{B_{\sigma}(p)} V(\phi) dV_g \right) \\ &\geq \int_{B_{\rho}(p) \setminus B_{\sigma}(p)} 4e^{a\tau^2} |\iota_{e_1} F_A|^2 dV_g + \int_{B_{\rho}(p) \setminus B_{\sigma}(p)} 2e^{a\tau^2} |D_{e_1}^{as} \phi|^2 dV_g \\ &- 2 \int_{\sigma}^{\rho} \frac{e^{a\tau^2}}{\tau} d\tau \int_{B_{\tau}(p)} |D_{e_1}^{as} \phi|^2 dV_g + n \int_{\sigma}^{\rho} \frac{e^{a\tau^2}}{\tau} d\tau \int_{B_{\tau}(p)} V(\phi) dV_g. \end{aligned} \quad (2.5.17)$$

Remark 3. *It is easy to see from Theorem (2.5.2) that if the Higgs energy ϕ and potential function V satisfy*

$$nV(\phi) - 2|D_{e_1}^{as} \phi|^2 \geq 0,$$

then the weighted energy functional $\mathcal{E}_{YMH}(\tau)$ is non-decreasing with respect to τ , and inequality (2.5.17) provides a lower bound estimate for the difference of $\mathcal{E}_{YMH}(\tau)$.

Part III: curvature estimate for YMH connection

Now we give a basic curvature for Yang-Mills-Higgs connection.

Chapter 3 Investigations into Instantons

1 Some geometric or Lie algebraic identities and inequalities

This section is devoted to developing several fundamental results concerning bundle-valued differential forms, such as key geometric and matrix-related identities and inequalities. These preliminary results provide the necessary groundwork for our study of instantons in the present chapter.

The Bochner-Weitzenböck formulas for Lie Algebra-bundle-valued forms

We first establish formulas relating the Hodge-deRham Laplacian and the rough Laplacian of $\mathfrak{g}_E$ -valued 1- and 2-forms. They involve both the curvature of the Riemannian base manifold and of the bundle with connection. We deduce from them an algebraic expression of the rough Laplacian of a harmonic $\mathfrak{g}_E$ -valued p -form ($p = 1$ or 2). These formulas are of fundamental importance in the theory.

To begin we define a basic (zero-order) operator $\mathfrak{R}^\nabla : \Omega^1(\mathfrak{g}_E) \rightarrow \Omega^1(\mathfrak{g}_E)$ by setting

$$\mathfrak{R}^\nabla(\varphi)_X \stackrel{\text{def}}{=} \sum_{j=1}^n [R_{e_j, X}^\nabla, \varphi_{e_j}],$$

where R^v denotes the curvature of the connection ∇ on E and $\{e_1, \dots, e_n\}$ is an orthonormal basis of the tangent space $T_x M$ at the point x . Recall that the Ricci transformation $\text{Ric} : T_x M \rightarrow T_x M$ is defined by

$$\text{Ric}(X) = \sum_{j=1}^n R_{X, e_j} e_j,$$

where R here denotes the Riemannian curvature tensor. Given φ in $\Omega^1(\mathfrak{g}_E)$ we define

$$(\varphi \circ \text{Ric})_X = \varphi_{\text{Ric}(X)}.$$

We recall that on the standard sphere $\text{Ric}(X) = (n - 1)X$.

Lemma 3.1. *For any φ in $\Omega^1(\mathfrak{g}_E)$, we have*

$$\Delta^\nabla \varphi = \nabla^* \nabla \varphi + \varphi \circ \text{Ric} + \mathfrak{R}^\nabla(\varphi). \quad (3.1)$$

Proof. Fix a point x in M and choose $X, e_1, \dots, e_n$ in $T_x M$ so that $\{e_1, \dots, e_n\}$ is an orthonormal basis. Extend X to a local vector field and $\{e_1, \dots, e_n\}$ to a local orthonormal frame field so that $(DX)(x) = (De_1)(x) = \dots = (De_n)(x) = 0$. Then, at the point x ,

$$(d^\nabla \delta^\nabla \varphi)_X = \nabla_X(\delta^\nabla \varphi) = -\nabla_X \left(\sum_{j=1}^n (\nabla_{e_j} \varphi) e_j \right) = -\sum_{j=1}^n (\nabla_{X, e_j}^2 \varphi) e_j, \quad (3.2)$$

and

$$\begin{aligned} (\delta^\nabla d^\nabla \varphi)_X &= -\sum_{j=1}^n (\nabla_{e_j} d^\nabla \varphi)_{e_j, X} = -\sum_{j=1}^n \nabla_{e_j} ((d^\nabla \varphi)_{e_j, X}) \\ &= -\sum_{j=1}^n \nabla_{e_j} ((\nabla_{e_j} \varphi)_X - (\nabla_X \varphi)_{e_j}) \\ &= -\sum_{j=1}^n \left((\nabla_{e_j, e_j}^2 \varphi)_X - (\nabla_{e_j, X}^2 \varphi)_{e_j} \right). \end{aligned} \quad (3.3)$$

Adding (3.1.2) and (3.1.3) we find that

$$(\Delta^\nabla \varphi)_X = (\nabla^* \nabla \varphi)_X + \sum_{j=1}^n (R_{e_j, X}^\nabla \varphi)_{e_j}, \quad (3.4)$$

here we keep the notation R^∇ for the curvature of the connection on the bundle $T^*M \otimes \mathfrak{g}_E$ induced by the Levi-Civita connection D on the base manifold and the connection V on the bundle E .

The operator R^∇ acts as a derivation, hence $(R_{X, Y}^\nabla \varphi)_Z = [R_{X, Y}^\nabla, \varphi_Z] - \varphi_{R_{X, Y} Z}$. Applying this formula to the second term in (3.1.4) yields

$$\begin{aligned} (\Delta^\nabla \varphi)_X &= (\nabla^* \nabla \varphi)_X + \sum_{j=1}^n \left([R_{e_j, X}^\nabla, \varphi_{e_j}] - \varphi_{R_{e_j, X} e_j} \right) \\ &= (\nabla^* \nabla \varphi)_X + \mathfrak{R}^\nabla(\varphi)_X + \varphi_{\text{Ric}(X)}. \end{aligned}$$

□

From the above lemma, we immediately deduce that if φ is a $\mathfrak{g}_E$ -valued ∇ -harmonic 1-form on the standard sphere S^n , then

$$\nabla^* \nabla \varphi = -(n-1)\varphi - \mathfrak{R}^\nabla(\varphi). \quad (3.5)$$

Similar calculations can be carried out for any p -form. Of particular interest here is the case $p = 2$. In analogy with before we define $\mathfrak{R}^\nabla : \Omega^2(\mathfrak{g}_E) \rightarrow \Omega^2(\mathfrak{g}_E)$ by

$$\mathfrak{R}^\nabla(\varphi)_{X,Y} = \sum_{j=1}^n \left([R_{e_j,X}^\nabla, \varphi_{e_j,Y}] - [R_{e_j,Y}^\nabla, \varphi_{e_j,X}] \right).$$

For a linear map ω on 2-vectors we define

$$(\varphi \circ \omega)_{X,Y} = \frac{1}{2} \sum_{j=1}^n \varphi_{e_j, \omega_{X,Y} e_j},$$

this is indeed the composition of ω and φ viewed as maps from $\Lambda^2 TM$ to $\mathfrak{g}_E$. Notice that with our convention $\varphi_{X,Y} = \varphi(X \wedge Y)$.

Notice then that the extension of the Ricci transformation to 2-forms as $\text{Ric} \wedge I$ is given by

$$(\text{Ric} \wedge I)_{X,Y} = \text{Ric}(X) \wedge Y + X \wedge \text{Ric}(Y).$$

After introducing the notation, we proceed to derive the Bochner-Weitzenböck formula for $\mathfrak{g}_E$ -valued 2-forms.

Lemma 3.2. *For any φ in $\Omega^2(\mathfrak{g}_E)$*

$$\Delta^\nabla \varphi = \nabla^* \nabla \varphi + \varphi \circ (\text{Ric} \wedge I + 2R) + \mathfrak{R}^\nabla(\varphi). \quad (3.6)$$

Proof. Fix a point x in M and choose $X, Y, e_1, \dots, e_n$ in $T_x M$ so that $\{e_1, \dots, e_n\}$ is an orthonormal basis. Extend X, Y to a local vector field and $\{e_1, \dots, e_n\}$ to a local orthonormal frame field so that $(DX)(x) = (DY)(x) = (De_1)(x) = \dots = (De_n)(x) = 0$. Then, at the point x ,

$$\begin{aligned} (d^\nabla \delta^\nabla \varphi)_{X,Y} &= (\nabla_X \delta^\nabla \varphi)_Y - (\nabla_Y \delta^\nabla \varphi)_X \\ &= -\nabla_X \left(\sum_{j=1}^n (\nabla_{e_j} \varphi)_{e_j, Y} \right) + \nabla_Y \left(\sum_{j=1}^n (\nabla_{e_j} \varphi)_{e_j, X} \right) \\ &= -\sum_{j=1}^n \left((\nabla_{X, e_j}^2 \varphi)_{e_j, Y} - (\nabla_{Y, e_j}^2 \varphi)_{e_j, X} \right). \end{aligned} \quad (3.7)$$

and

$$\begin{aligned} (\delta^\nabla d^\nabla \varphi)_{X,Y} &= -\sum_{j=1}^n (\nabla_{e_j} d^\nabla \varphi)_{e_j, X, Y} \\ &= -\sum_{j=1}^n \nabla_{e_j} \left((\nabla_{e_j} \varphi)_{X, Y} + (\nabla_Y \varphi)_{e_j, X} - (\nabla_X \varphi)_{e_j, Y} \right) \\ &= -\sum_{j=1}^n \left((\nabla_{e_j, e_j}^2 \varphi)_{X, Y} + (\nabla_{e_j, Y}^2 \varphi)_{e_j, X} - (\nabla_{e_j, X}^2 \varphi)_{e_j, Y} \right). \end{aligned} \quad (3.8)$$

Combining (3.1.7) and (3.1.8) we obtain

$$(\Delta^\nabla \varphi)_{X,Y} = (\nabla^* \nabla \varphi)_{X,Y} + \sum_{j=1}^n \left((R_{e_j,X}^\nabla \varphi)_{e_j,Y} - (R_{e_j,Y}^\nabla \varphi)_{e_j,X} \right). \quad (3.9)$$

By using the identity

$$(R_{X,Y}^\nabla \varphi)_{Z,W} = [R_{X,Y}^\nabla, \varphi_{Z,W}] - \varphi_{R_{X,Y}Z,W} - \varphi_{Z,R_{X,Y}W},$$

Equation (3.1.9) can be expressed as

$$\begin{aligned} (\Delta^\nabla \varphi)_{X,Y} &= (\nabla^* \nabla \varphi)_{X,Y} + \sum_{j=1}^n \left([R_{e_j,X}^\nabla, \varphi_{e_j,Y}] - \varphi_{R_{e_j,X}e_j,Y} - \varphi_{e_j,R_{e_j,X}Y} \right) \\ &\quad - \sum_{j=1}^n \left([R_{e_j,Y}^\nabla, \varphi_{e_j,X}] - \varphi_{R_{e_j,Y}e_j,X} - \varphi_{e_j,R_{e_j,Y}X} \right) \\ &= (\nabla^* \nabla \varphi)_{X,Y} + \mathfrak{R}^\nabla(\varphi)_{X,Y} \\ &\quad + \underbrace{\varphi_{\text{Ric}(X),Y} - \sum_{j=1}^n \varphi_{e_j,R_{e_j,X}Y} - \varphi_{\text{Ric}(Y),X} + \sum_{j=1}^n \varphi_{e_j,R_{e_j,Y}X}}_{\text{remainder term}}. \end{aligned}$$

Then we use the first Bianchi identity

$$\begin{aligned} R_{e_j,X}Y &= -R_{X,Y}e_j - R_{Y,e_j}X \\ R_{e_j,Y}X &= -R_{Y,X}e_j - R_{X,e_j}Y \end{aligned}$$

to simplify the remainder terms, we obtain

$$\begin{aligned} \text{remainder term} &= \varphi_{(\text{Ric} \wedge I)(X,Y)} + \sum_{j=1}^n \varphi_{e_j, R_{X,Y}e_j} \\ &= (\varphi \circ (\text{Ric} \wedge I))_{X,Y} + 2(\varphi \circ R)_{X,Y} \end{aligned}$$

We finally arrive at the Weitzenböck formula (3.1.6) for 2-forms. □

From the above lemma, we immediately deduce that if φ is a $\mathfrak{g}_E$ -valued harmonic 2-form on the standard sphere S^n , then

$$\nabla^* \nabla \varphi = -2(n-2)\varphi - \mathfrak{R}^v(\varphi). \quad (3.10)$$

Lemma 3.3. *Let (M^4, g) be a Riemannian manifold, and suppose $E \rightarrow M$ is a smooth vector bundle with connection ∇ . If $\omega \in \Lambda_+^2(\mathfrak{g}_E)$ is a harmonic two-form, one has*

$$\frac{1}{2} \Delta |\omega|^2 = |\nabla \omega|^2 - \langle \omega, [F^+, \omega] \rangle - 2 \langle \omega, W^+ \star \omega \rangle + \frac{1}{3} R |\omega|^2, \quad (3.11)$$

where with respect to local bases one has, for $P, Q \in \Lambda_+^2(\mathfrak{g}_E)$, the tensor $[P, Q] \in \Lambda_+^2(\mathfrak{g}_E)$ defined via

$$[P, Q]_{ij\alpha}^\beta := g^{kl} \left(P_{ik\delta}^\beta Q_{jl\alpha}^\delta - P_{ik\alpha}^\delta Q_{jl\delta}^\beta - P_{jk\delta}^\beta Q_{il\alpha}^\delta + P_{jk\alpha}^\delta Q_{il\delta}^\beta \right),$$

and

$$(W^+ \star \omega)_{ij\alpha}^\beta := g^{kp} g^{lq} W_{ijkl}^+ \omega_{pq\alpha}^\beta.$$

Proof. □

Some sharp matrix inequalities

Lemma 3.4. *Let (M^4, g) be a Riemannian manifold, and suppose $E \rightarrow M$ is a smooth vector bundle with connection ∇ . If $\omega \in \Lambda_+^2(\mathfrak{g}_E)$ is a two-form, one has*

$$|\langle \omega, W^+ \star \omega \rangle| \leq \frac{2}{\sqrt{6}} |W^+| |\omega|^2. \quad (3.12)$$

Proof. 1 □

Definition 3.5. *Given $\mathfrak{g} \subset \mathfrak{so}E$, let*

$$\begin{aligned} \gamma_0 &:= \sup_{A, B \in \mathfrak{g} \setminus \{0\}} \frac{|[A, B]|}{|A||B|}, \\ \gamma_1 &:= \sup_{\omega \in \Lambda_+^2(\mathfrak{g}) \setminus \{0\}} \frac{\langle \omega, [\omega, \omega] \rangle}{|\omega|^3}. \end{aligned}$$

Lemma 3.6. *Given $\mathfrak{g} \subset \mathfrak{so}E$, one has $\gamma_0 \leq \sqrt{2}$, with the supremum defining γ_0 attained by pairs A, B which are simultaneously equivalent to a pair of Pauli matrices, i.e.*

$$A = \left(\begin{array}{cc|cc} 0 & t & & \\ -t & 0 & & \\ \hline & & 0 & 0 \\ 0 & & 0 & t \\ \hline & & -t & 0 \\ 0 & & 0 & 0 \end{array} \right), \quad B = \left(\begin{array}{cc|cc} & & -s & 0 \\ & & 0 & s \\ \hline s & 0 & & \\ 0 & -s & & 0 \\ \hline & & 0 & 0 \\ 0 & & 0 & 0 \end{array} \right). \quad (3.13)$$

Furthermore, in the case $\mathfrak{g} = \mathfrak{so}_3$ one has $\gamma_0 = 1$.

Proof. We wish to maximize

$$F(A, B) = \|[A, B]\|^2$$

subject to the constraints $\|A\|^2 = 1$ and $\|B\|^2 = 1$. Introduce Lagrange multipliers $\lambda, \mu \in \mathbb{R}$ and consider the functional

$$\Phi(A, B, \lambda, \mu) = \|[A, B]\|^2 - \lambda(\|A\|^2 - 1) - \mu(\|B\|^2 - 1).$$

We look for stationary points of Φ under arbitrary variations $A \mapsto A + \varepsilon H$, $B \mapsto B + \varepsilon K$ with $H, K \in \mathfrak{g}$.

- Variation in A :

Write $A_\varepsilon = A + \varepsilon H$, then

$$\begin{aligned}\| [A_\varepsilon, B] \|^2 &= \langle [A, B] + \varepsilon [H, B], [A, B] + \varepsilon [H, B] \rangle \\ &= \| [A, B] \|^2 + 2\varepsilon \langle [H, B], [A, B] \rangle + O(\varepsilon^2).\end{aligned}$$

Hence the first-order variation is

$$\delta_A \| [A, B] \|^2 (H) = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \| [A_\varepsilon, B] \|^2 = 2 \langle [H, B], [A, B] \rangle.$$

Notice the invariance of the trace inner product, i.e.

$$\begin{aligned}\langle [H, B], [A, B] \rangle &= -\frac{1}{2} \text{Tr}((HB - BH)(AB - BA)) \\ &= -\frac{1}{2} (\text{Tr}(HBAB) - \text{Tr}(HB^2A) - \text{Tr}(BHAB) + \text{Tr}(BHBA)) \\ &= -\frac{1}{2} (\text{Tr}(HBAB) - \text{Tr}(HB^2A) - \text{Tr}(HAB^2) + \text{Tr}(HBAB)) \\ &= -\frac{1}{2} \text{Tr}(H(B(AB - BA) - (AB - BA)B)) \\ &= \langle H, [B, [A, B]] \rangle \\ &= -\langle H, (\text{ad}B)^2(A) \rangle,\end{aligned}$$

and similarly $\delta_A (\|A\|^2)(H) = 2\langle H, A \rangle$, the Lagrange condition $\delta_A \Phi = 0$ for all H yields

$$(\text{ad}B)^2(A) + \lambda A = 0, \quad \text{i.e. } (\text{ad}B)^2 A = -\lambda A.$$

- Variation in B :

A perfectly analogous computation fixing A and varying $B \mapsto B + \varepsilon K$ gives $\delta_B \Phi = 0 \forall K$ and hence

$$(\text{ad}A)^2 B = -\mu B.$$

One also checks by taking inner product of each with A or B respectively that $\lambda = \mu = \| [A, B] \|^2$.

Since B is skew-symmetric, there exists an orthonormal basis in which B splits into 2×2 blocks of the form $\begin{pmatrix} 0 & b_i \\ -b_i & 0 \end{pmatrix}$ plus zeros. The eigen-equation $(\text{ad}B)^2 A = -\| [A, B] \|^2 A$ forces A to have nonzero entries only in two distinct such 2×2 blocks. One checks directly that choosing exactly two blocks of size 2 with nonzero scalars t, s reproduces the eigen-equations:

$$A \sim t \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus t \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad B \sim s \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus (-s) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

A direct norm computation gives $\|A\|^2 = 2t^2$, $\|B\|^2 = 2s^2$, $\|[A, B]\|^2 = 8s^2t^2$, and hence

$$\frac{\|[A, B]\|}{\|A\|\|B\|} = \frac{\sqrt{8s^2t^2}}{\sqrt{2t^2}\sqrt{2s^2}} = \sqrt{2}.$$

This shows $\gamma_0 \leq \sqrt{2}$ and that equality is attained precisely by the Pauli-type blocks above.

Eventually, we consider the case $\mathfrak{so}_3$. We equip $\mathfrak{so}_3$ with the inner product

$$\langle X, Y \rangle = -\frac{1}{2} \text{Tr}(XY), \quad \|X\|^2 = \langle X, X \rangle,$$

so that the identification

$$a = (a_1, a_2, a_3) \in \mathbb{R}^3 \longleftrightarrow A = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix} \in \mathfrak{so}_3$$

becomes an *isometry*: $\|A\| = \|a\|$. Under this identification one checks immediately that

$$[A, B] \longleftrightarrow a \times b, \quad \text{and hence} \quad \|[A, B]\| = \|a \times b\| \leq \|a\| \|b\|.$$

Therefore for unit-norm matrices $\|A\| = \|B\| = 1$ we get

$$\frac{\|[A, B]\|}{\|A\| \|B\|} = \frac{\|a \times b\|}{\|a\| \|b\|} \leq 1.$$

Thus in the special case $\mathfrak{g} = \mathfrak{so}_3$ one indeed has

$$\gamma_0 = \sup_{\|A\|=\|B\|=1} \frac{\|[A, B]\|}{\|A\| \|B\|} = 1.$$

□

Next, using Lemma 3.1.6, we establish an estimate for the wedge product commutator of Lie algebra valued 2-forms. We first record an elementary linear algebra fact. Note here that we take the metric induced on two-forms as

$$\langle \eta, \mu \rangle := g^{ik} g^{jl} \eta_{ij} \mu_{kl}.$$

Lemma 3.7. *Given (V^4, g) an inner product space, for $\eta, \mu \in \Lambda_+^2(V^*)$ define $(\eta \circ \mu) \in \Lambda_+^2(V^*)$ via*

$$(\eta \circ \mu)_{ij} = g^{kl} (\eta_{ik} \mu_{jl} - \eta_{jk} \mu_{il}).$$

Then given $\{e_1, e_2, e_3\}$ an orthonormal basis for $\Lambda_+^2(V^)$, one has that $\{(e_1 \circ e_2), (e_1 \circ e_3), (e_2 \circ e_3)\}$ is another orthonormal basis.*

Proof. Since V is a four-dimensional inner product space, choose an orthonormal basis $\{e_1, e_2, e_3, e_4\}$ for V such that the components of the metric g satisfy $g_{ij} = \delta_{ij}$ and $g^{ij} = \delta^{ij}$. In $\Lambda_+^2(V^*)$, an orthonormal basis can be defined as:

$$\omega_1 = e^1 \wedge e^2 + e^3 \wedge e^4, \quad \omega_2 = e^1 \wedge e^3 - e^2 \wedge e^4, \quad \omega_3 = e^1 \wedge e^4 + e^2 \wedge e^3,$$

where $\{e^i\}$ are the dual bases. Calculating the inner product:

$$\langle \omega_i, \omega_j \rangle = g^{pk} g^{ql} (\omega_i)_{pq} (\omega_j)_{kl} = \sum_{p,q} (\omega_i)_{pq} (\omega_j)_{pq} = 4\delta_{ij},$$

therefore, after normalization, the orthonormal basis for $\Lambda_+^2(V^*)$ becomes:

$$e_a = \frac{1}{2}\omega_a, \quad a = 1, 2, 3,$$

satisfying $\langle e_a, e_b \rangle = \delta_{ab}$.

Using the non-normalized ω_a to calculate $\omega_p \circ \omega_q$,

$$(\omega_1 \circ \omega_2)_{ij} = \sum_{k=1}^4 [(\omega_1)_{ik} (\omega_2)_{jk} - (\omega_1)_{jk} (\omega_2)_{ik}].$$

Non-zero components: $(\omega_1 \circ \omega_2)_{14} = 2$, $(\omega_1 \circ \omega_2)_{41} = -2$, $(\omega_1 \circ \omega_2)_{23} = 2$, $(\omega_1 \circ \omega_2)_{32} = -2$. Comparing with $\omega_3 = e^1 \wedge e^4 + e^2 \wedge e^3$, we have

$$\omega_1 \circ \omega_2 = 2\omega_3.$$

Similar calculations follow for $\omega_1 \circ \omega_3$ and $\omega_2 \circ \omega_3$. Then via the normalized bases $e_a = \frac{1}{2}\omega_a$, we obtain

$$e_1 \circ e_2 = \left(\frac{1}{2}\omega_1\right) \circ \left(\frac{1}{2}\omega_2\right) = \frac{1}{4}(\omega_1 \circ \omega_2) = \frac{1}{4} \cdot 2\omega_3 = e_3.$$

Similarly for $e_1 \circ e_3$ and $e_2 \circ e_3$. Hence, we deduce

$$\{e_1 \circ e_2, e_1 \circ e_3, e_2 \circ e_3\} = \{e_3, -e_2, e_1\}.$$

This completes the proof of the lemma. □

Lemma 3.8. *Let $E \rightarrow (M^4, g)$ be a smooth vector bundle. Given $P \in \Lambda_+^2(\mathfrak{g}_E)$, one has*

$$\|[P, P]\| \leq \frac{2}{\sqrt{3}} \gamma_0 |P|^2. \quad (3.14)$$

Proof. Fix a point $p \in M$ and let $\{e_i\}$, $i = 1, 2, 3$ be an orthonormal basis for Λ_+^2 at p . Now let us express $P = P^i e_i$, where $P^i \in \mathfrak{g}_E$. Due to the skew commutativity of the bracket structure we may estimate

$$\begin{aligned} \| [P, P] \|^2 &= \| 2(e_1 \circ e_2) \otimes [P^1, P^2] + 2(e_1 \circ e_3) \otimes [P^1, P^3] + 2(e_2 \circ e_3) \otimes [P^2, P^3] \|^2 \\ &\leq 4 \left(\| [P^1, P^2] \|^2 + \| [P^1, P^3] \|^2 + \| [P^2, P^3] \|^2 \right) \\ &\leq 4\gamma_0^2 \left(|P^1|^2 |P^2|^2 + |P^1|^2 |P^3|^2 + |P^2|^2 |P^3|^2 \right) \\ &\leq \frac{4\gamma_0^2}{3} \left(|P^1|^2 + |P^2|^2 + |P^3|^2 \right)^2 \\ &= \frac{4\gamma_0^2}{3} |P|^4. \end{aligned}$$

Taking the square root yields the claim. □

Improved Kato inequality

Proposition 3.9. *Let $E \rightarrow (M^4, g)$ be a vector bundle over a smooth Riemannian four-manifold. Given ∇ a connection on E , and $\omega \in \Lambda^2(\mathfrak{g}_E)$ a harmonic two-form, one has the pointwise inequality*

$$|\nabla \omega|^2 \geq \frac{3}{2} |d|\omega||^2. \quad (3.15)$$

Proof. First, by applying the Bochner formula (Lemma 3.1.3) to ω (summing over self-dual and antiself-dual parts) we obtain

$$\frac{1}{2} \Delta |\omega|^2 = |\nabla \omega|^2 + \left\langle [F, \omega] - 2W \star \omega + \frac{R}{3} \omega, \omega \right\rangle. \quad (3.16)$$

We now make a conformal modification of the metric and derive a second Böchner identity. In particular, let $\hat{g} = |\omega|g$, which defines a smooth Riemannian metric away from the zero locus of $|\omega|$. Note by construction that $|\omega|_{\hat{g}}^2 \equiv 1$. Furthermore, as the condition that ω is harmonic is conformally invariant, ω is harmonic with respect to $\hat{g}$, and thus we apply the Bochner formula again to conclude

$$0 = \frac{1}{2} \hat{\Delta} |\omega|_{\hat{g}}^2 = |\hat{\nabla} \omega|_{\hat{g}}^2 + \left\langle [\hat{F}, \omega] - 2\hat{W} \star \omega + \frac{\hat{R}}{3} \omega, \omega \right\rangle_{\hat{g}}. \quad (3.17)$$

The bundle curvature F and the Weyl tensor W are conformally covariant, yielding

$$\left\langle [\hat{F}, \omega] - 2\hat{W} \star \omega, \omega \right\rangle_{\hat{g}} = |\omega|^{-3} \langle [F, \omega] - 2W \star \omega, \omega \rangle. \quad (3.18)$$

On the other hand, using the transformation formula for the scalar curvature under conformal change one obtains

$$\left\langle \frac{\hat{R}}{3}\omega, \omega \right\rangle_{\hat{g}} = |\omega|^{-3} \left(\left\langle \frac{R}{3}\omega, \omega \right\rangle - \frac{1}{2}\Delta|\omega|^2 + \frac{3}{2}|d|\omega||^2 \right). \quad (3.19)$$

Plugging (3.1.18) and (3.1.19) into (3.1.17), and incorporating (3.1.16) we conclude

$$\begin{aligned} |\hat{\nabla}\omega|_{\hat{g}}^2 &= -|\omega|^{-3} \left(\langle [F, \omega] + W \star \omega + \frac{R}{6}\omega, \omega \rangle - \frac{1}{2}\Delta|\omega|^2 + \frac{3}{2}|d|\omega||^2 \right) \\ &= |\omega|^{-3} \left(|\nabla\omega|^2 - \frac{3}{2}|d|\omega||^2 \right). \end{aligned}$$

This implies the desired inequality away from the vanishing locus of $|\omega|$, which in turn implies the inequality at all points. □

2 A Conformally Invariant Gap Theorem in Yang-Mills Theory

Introduction and Motivation

In this section, we will explore a significant advance in the theory of Yang–Mills connections over four-dimensional manifolds, focusing on a conformally invariant energy gap theorem. Yang–Mills theory in four dimensions plays a central role in both mathematics and physics. Critical points of the Yang–Mills energy functional,

$$E(\nabla) = \int_X |F_{\nabla}|^2 dV_g,$$

are called Yang–Mills connections. Among these, instantons—solutions with self-dual or anti-self-dual curvature—are energy minimizers and foundational objects in Donaldson theory and four-manifold topology.

Despite their centrality, not all Yang–Mills connections are instantons. A fundamental and long-standing question is whether one can understand the energy spectrum of Yang–Mills connections beyond the self-dual sector. In particular, if a connection is not an instanton, can we say something quantitative about how much higher its energy must be?

Previous works have established “gap theorems” under curvature positivity assumptions or ε -regularity frameworks. However, these conditions are restrictive. The main contribution of Gursky, Kelleher, and Streets is to lift these constraints and provide a

sharp, conformally invariant energy lower bound for non-self-dual Yang–Mills connections when the underlying manifold has positive Yamabe invariant.

The central result of the paper is a dichotomy: for any Yang–Mills connection on a closed, oriented four-manifold, either the connection is anti-self-dual, or it must obey a sharp lower bound involving the Yamabe invariant and the L^2 norms of the curvature and Weyl tensor.

Theorem 3.10 (The Main Theorem). *Let (X^4, g) be a closed, oriented four-manifold. Suppose ∇ is a Yang–Mills connection on a vector bundle E over X^4 with structure group $G \subset SO(E)$, and curvature F_∇ . Then one of the following must hold:*

- $F_\nabla^+ \equiv 0$; or
- F_∇^+ satisfies

$$Y([g]) \leq 3\gamma_1 \|F_\nabla^+\|_{L^2} + 2\sqrt{6} \|W^+\|_{L^2}, \quad (3.20)$$

where $\gamma_1 = \gamma_1(E) \leq \frac{4}{\sqrt{6}}$ is a constant which depends on the structure group of the bundle, and W^+ is the self-dual Weyl tensor.

Moreover, if equality holds in (1.1) then $[g]$ admits a Yamabe metric $\bar{g}$ with respect to which W^+ has constant norm, $\nabla_{\bar{g}} F_{\nabla}^+ \equiv 0$, and

$$R_{\bar{g}} - 2\sqrt{6} |W^+|_{\bar{g}} = 3\gamma_1 |F_\nabla^+|_{\bar{g}}, \quad (3.21)$$

where $R_{\bar{g}}$ is the scalar curvature of $\bar{g}$. Furthermore, if $\gamma_1 > 0$ we must have $b_2^+(M^4) = 0$, whereas if $\gamma_1 = 0$ then all harmonic self-dual forms are parallel.

Overview of the Proof

The central innovation of the proof lies in reducing a Yang–Mills energy gap problem to a conformally invariant variational problem of Yamabe type.

The argument begins with a Bochner formula applied to the self-dual part F_∇^+ of the curvature of a Yang–Mills connection. Because F^+ is harmonic, it satisfies a Weitzenböck-type identity that includes terms involving $|\nabla F^+|^2$, the Weyl curvature W^+ , and a non-linear Lie bracket $[F^+, F^+]$. These latter two terms are estimated using sharp algebraic inequalities, introducing a key constant γ_1 that depends only on the structure group.

Next, an improved Kato inequality for F^+ is applied, allowing the authors to transform the Bochner identity into a differential inequality for $|F^+|^{1/2}$. This inequality takes the schematic form:

$$L(|F^+|^{1/2}) \leq 0,$$

where L is a modified Yamabe-type operator. Here lies the key insight: this operator behaves like the classical conformal Laplacian and retains the same conformal covariance. Thus, the authors reinterpret the problem of bounding the Yang–Mills energy as a spectral problem for L , analogous to the Yamabe problem of finding constant scalar curvature metrics in a conformal class.

By testing L against suitable test functions (in particular, $|F^+|^{1/2}$ and the first eigenfunction of L), one obtains an integral inequality. Combining this with the variational characterization of the Yamabe invariant yields the estimate in the main theorem.

Thus, a sharp energy gap for non-self-dual Yang–Mills connections is derived by embedding the curvature term into a conformally covariant scalar curvature problem. This elegant translation from gauge theory to conformal geometry is the conceptual heart of the proof.

Preparatory Ingredients

Bochner-Weitzenböck Formula for Self-Dual Forms

Let (X^4, g) be a Riemannian manifold, and suppose $E \rightarrow X$ is a smooth vector bundle with connection ∇ . If $\omega \in \Lambda_+^2(\mathfrak{g}_E)$ is a harmonic two-form, one has

$$\frac{1}{2}\Delta|\omega|^2 = |\nabla\omega|^2 - \langle\omega, [F^+, \omega]\rangle - 2\langle\omega, W^+ \star \omega\rangle + \frac{1}{3}R|\omega|^2, \quad (3.22)$$

where with respect to local bases one has, for $P, Q \in \Lambda_+^2(\mathfrak{g}_E)$, the tensor $[P, Q] \in \Lambda_+^2(\mathfrak{g}_E)$ defined via

$$[P, Q]_{ij\alpha}^\beta := g^{kl} \left(P_{ik\delta}^\beta Q_{jl\alpha}^\delta - P_{ik\alpha}^\delta Q_{jl\delta}^\beta - P_{jk\delta}^\beta Q_{il\alpha}^\delta + P_{jk\alpha}^\delta Q_{il\delta}^\beta \right),$$

and

$$(W^+ \star \omega)_{ij\alpha}^\beta := g^{kp} g^{lq} W_{ijkl}^+ \omega_{pq\alpha}^\beta.$$

For the self-dual part of the curvature 2-form of a Yang–Mills connection F_∇^+ , since it is harmonic (i.e., both $dF^+ = 0$ and $d^*F^+ = 0$), the Bochner-Weitzenböck formula for bundle-valued self-dual 2-forms applies:

$$\frac{1}{2}\Delta|F^+|^2 = |\nabla F^+|^2 - \langle F^+, [F^+, F^+]\rangle - 2\langle F^+, W^+ \star F^+\rangle + \frac{1}{3}R|F^+|^2. \quad (3.23)$$

This forms the analytic starting point of the proof. Each term will be carefully bounded or interpreted geometrically.

Estimate for the Weyl Tensor Term

Let (M^4, g) be a Riemannian manifold, and suppose $E \rightarrow M$ is a smooth vector bundle with connection ∇ . If $\omega \in \Lambda_+^2(\mathfrak{g}_E)$ is a two-form, one has

$$|\langle \omega, W^+ \star \omega \rangle| \leq \frac{2}{\sqrt{6}} |W^+| |\omega|^2. \quad (3.24)$$

Thus the action of the self-dual Weyl curvature W^+ on F_∇^+ is

$$|\langle F^+, W^+ \star F^+ \rangle| \leq \frac{2}{\sqrt{6}} |W^+| |F^+|^2.$$

This allows one to extract a bound involving the L^2 -norm of the Weyl tensor in the eventual inequality.

Lie-Algebraic Estimate

The nonlinearity $\langle F^+, [F^+, F^+] \rangle$ arises from the curvature bracket. To control this term, one defines two universal constants depending on the Lie algebra $\mathfrak{g} \subset \mathfrak{so}_E$:

$$\begin{aligned} \gamma_0 &:= \sup_{A, B \in \mathfrak{g} \setminus \{0\}} \frac{|[A, B]|}{|A||B|}, \\ \gamma_1 &:= \sup_{\omega \in \Lambda_+^2(\mathfrak{g}) \setminus \{0\}} \frac{\langle \omega, [\omega, \omega] \rangle}{|\omega|^3}. \end{aligned}$$

On the one side, $\gamma_0 \leq \sqrt{2}$, with the supremum defining γ_0 attained by pairs A, B which are simultaneously equivalent to a pair of Pauli matrices, i.e.

$$A = \left(\begin{array}{cc|cc} 0 & t & 0 & 0 \\ -t & 0 & 0 & 0 \\ \hline 0 & 0 & t & 0 \\ 0 & -t & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right), \quad B = \left(\begin{array}{cc|cc} 0 & -s & 0 & 0 \\ 0 & 0 & s & 0 \\ \hline s & 0 & 0 & 0 \\ 0 & -s & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right).$$

In the case $\mathfrak{g} = \mathfrak{so}_3$ one has $\gamma_0 = 1$.

On the other side, for $P \in \Lambda_+^2(\mathfrak{g}_E)$, one has

$$\|[P, P]\| \leq \frac{2}{\sqrt{3}} \gamma_0 |P|^2. \quad (3.25)$$

Improved Kato Inequality

For harmonic 2-forms valued in the Lie algebra bundle, one has the sharp Kato-type inequality:

$$|\nabla F^+|^2 \geq \frac{3}{2} |d|F^+|^2. \quad (3.26)$$

The Classical and Modified Yamabe Problem

The classical Yamabe problem seeks to find a conformal metric $g' = u^2g$ on X^4 such that $R_{g'} = \text{const.}$ The key tool in this setting is the conformal Laplacian (or Yamabe operator):

$$L_g := -6\Delta + R,$$

which satisfies the conformal covariance:

$$L_{g'}(f) = u^{-3}L_g(uf), \quad \text{for } g' = u^2g.$$

The sign of the first eigenvalue of L_g :

$$\lambda_1(L_g) = \inf_{\phi \in C^\infty, \phi \neq 0} \frac{\int_X \phi L_g \phi dV_g}{\int_X \phi^2 dV_g}.$$

is a conformal invariant and determines the sign of scalar curvature realizable within $[g]$.

Modified Yamabe Problem and Conformal Covariance

By applying the Bochner–Weitzenböck formula (3.2.4), along with the estimate for the Weyl tensor term and the definition of γ_1 , we arrive at

$$\frac{1}{2}\Delta|F^+|^2 \geq |\nabla F^+|^2 + \frac{1}{3}(R - 2\sqrt{6}|W^+| - 3\gamma_1|F^+|)|F^+|^2. \quad (3.27)$$

By the Leibniz rule, away from the zero locus of F^+ we have

$$\frac{1}{2}\Delta|F^+|^2 = |F^+|\Delta|F^+| + |\nabla|F^+||^2,$$

hence by Improved Kato inequality (3.2.7),

$$\Delta|F^+| \geq \frac{1}{2} \frac{|\nabla|F^+||^2}{|F^+|} + \frac{1}{3}(R - 2\sqrt{6}|W^+| - 3\gamma_1|F^+|)|F^+|.$$

It follows that

$$\Delta|F^+|^{1/2} \geq \frac{1}{6}(R - 2\sqrt{6}|W^+| - 3\gamma_1|F^+|)|F^+|^{1/2} \quad (3.28)$$

off the zero locus of F^+ , or in the sense of distributions.

We next exploit (3.2.9) in conjunction with a modified Yamabe problem. In particular, given a metric $\hat{g} \in [g]$ we define the **modified scalar curvature**:

$$\Phi_{\hat{g}} = R_{\hat{g}} - 2\sqrt{6}|W^+|_{\hat{g}} - 3\gamma_1|F^+|_{\hat{g}}.$$

If we drop the Weyl and F^+ -terms then Φ is just the scalar curvature. Moreover, because these terms transform by scalings of the same weight under conformal changes

of metric, the transformation law for Φ is essentially the same as the scalar curvature. More precisely, if we define the natural generalization of the conformal Laplacian, i.e. **modified Yamabe operator** by

$$L = -6\Delta + \Phi,$$

then given $\hat{g} = u^2g$ it follows that

$$\Phi_{\hat{g}} = u^{-3}L_g u. \quad (3.29)$$

The verification of this formula is as follows.

In dimension 4, scalar curvature transforms under a conformal change $\hat{g} = u^2g$ as

$$R_{\hat{g}} = u^{-2} (R_g - 6u^{-1}\Delta_g u) = u^{-3} (-6\Delta_g u + R_g u).$$

Under conformal change, $|W^+|$ rescales as a 2-form:

$$|W^+|_{\hat{g}} = u^{-2}|W^+|_g.$$

Similarly, since F^+ is a self-dual 2-form with values in $\mathfrak{g}_E$, its norm transforms as:

$$|F^+|_{\hat{g}} = u^{-2}|F^+|_g.$$

Putting together the above transformations:

$$\begin{aligned} \Phi_{\hat{g}} &= R_{\hat{g}} - 2\sqrt{6}|W^+|_{\hat{g}} - 3\gamma_1|F^+|_{\hat{g}} \\ &= u^{-3} (-6\Delta_g u + R_g u) - 2\sqrt{6}u^{-2}|W^+|_g - 3\gamma_1 u^{-2}|F^+|_g \\ &= u^{-3} \left(-6\Delta_g u + R_g u - 2\sqrt{6}|W^+|_g u - 3\gamma_1|F^+|_g u \right) \\ &= u^{-3} (-6\Delta_g u + \Phi_g u) = u^{-3}L_g u. \end{aligned}$$

Moreover, L is conformally covariant:

$$L_{\hat{g}}\phi = u^{-3}L_g(u\phi). \quad (3.30)$$

The verification of this formula is as follows.

From the known transformation of the Laplacian on scalar functions in dimension 4, we have

$$-6\Delta_{\hat{g}}\phi = -6u^{-2} (\Delta_g \phi + 2u^{-1}\langle du, d\phi \rangle_g).$$

Next compute $L_g(u\phi)$,

$$\begin{aligned} L_g(u\phi) &= -6\Delta_g(u\phi) + \Phi_g u\phi \\ &= -6(\phi\Delta_g u + u\Delta_g \phi + 2\langle du, d\phi \rangle) + \Phi_g u\phi. \end{aligned}$$

Then

$$\begin{aligned}
 u^{-3}L_g(u\phi) &= u^{-3}[-6\phi\Delta_g u - 6u\Delta_g \phi - 12\langle du, d\phi \rangle + \Phi_g u\phi] \\
 &= -6u^{-2}(\Delta_g \phi + 2u^{-1}\langle du, d\phi \rangle) + \phi u^{-3}(-6\Delta_g u + \Phi_g u) \\
 &= -6\Delta_{\hat{g}}\phi + \phi\Phi_{\hat{g}} = L_{\hat{g}}\phi.
 \end{aligned}$$

Consequently, the sign of $\lambda_1(g)$ is a conformal invariant. In particular, by using an eigenfunction associated with $\lambda_1(L)$ as a conformal factor, it follows that $[g]$ admits a metric $\hat{g}$ with $\Phi_{\hat{g}} > 0$ (resp., $= 0$, < 0) if and only if $\lambda_1(L_g) > 0$ (resp., $= 0$, < 0).

One departure from the classical Yamabe problem is that the modified scalar curvature may only be Lipschitz continuous. Therefore, the Schauder estimates imply that the first eigenfunction is in $C^{2,\alpha}$ and hence defines a conformal metric which is only $C^{2,\alpha}$. One can smooth $|F^+|$ and approximate, but in our setting this will not be necessary.

Now suppose that $F^+ \not\equiv 0$. Then from the inequality (3.2.9), $u^{1/2}$ satisfies

$$0 \geq L_g(|F^+|^{1/2}).$$

Multiplying by $|F^+|^{1/2}$ and integrating over M^4 gives

$$0 \geq \int_X |F^+|^{1/2} L_g(|F^+|^{1/2}) dV_g.$$

so the lowest eigenvalue $\lambda_1(L) \leq 0$.

Let $\phi > 0$ be the eigenfunction associated with λ_1 , and define $g' := \phi^2 g$. Then:

$$\Phi_{g'} = \phi^{-3} L\phi = \lambda_1 \phi^{-2} \leq 0.$$

Hence, in this conformal metric g' , the modified scalar curvature is nonpositive:

$$R_{g'} - 2\sqrt{6}|W^+|_{g'} - 3\gamma_1|F^+|_{g'} \leq 0.$$

Integrating over X yields

$$\int_X R_{g'} dV_{g'} \leq 2\sqrt{6} \int_X |W^+|_{g'} dV_{g'} + 3\gamma_1 \int_X |F^+|_{g'} dV_{g'}.$$

Using the Cauchy–Schwarz inequality and conformal invariance of the L^2 norms:

$$\int_X R_{g'} dV_{g'} \leq \left(2\sqrt{6}\|W^+\|_{L^2} + 3\gamma_1\|F^+\|_{L^2}\right) \cdot \text{Vol}(g')^{1/2}.$$

On the other hand, by the definition of the Yamabe invariant,

$$\int_X R_{g'} dV_{g'} \geq Y([g]) \cdot \text{Vol}(g')^{1/2}.$$

Combining these facts, we obtain

$$Y([g]) \leq 3\gamma_1\|F^+\|_{L^2} + 2\sqrt{6}\|W^+\|_{L^2},$$

which is precisely the inequality (3.2.1) claimed in the main theorem.

Equality Case and Geometric Consequences

Suppose equality holds in the main inequality:

$$Y([g]) = 3\gamma_1 \|F_{\nabla}^+\|_{L^2} + 2\sqrt{6} \|W^+\|_{L^2}.$$

We now explain why this leads to strong geometric consequences.

Step 1: Consequences of Cauchy–Schwarz equality

Equality in the Cauchy–Schwarz inequality implies that the functions involved must be proportional to constants:

$$|F^+| = \text{const}, \quad |W^+| = \text{const},$$

in the Yamabe-minimizing conformal metric g' . That is, the pointwise norms of the self-dual curvature and the Weyl tensor are constant over X^4 with respect to g' .

Step 2: Vanishing of ∇F^+

Returning to the Bochner formula in the Yamabe metric g' , and using that all scalar functions are constant, the inequality becomes:

$$\frac{1}{2} \Delta |F^+|^2 = |\nabla F^+|^2 \geq 0.$$

But $|F^+|$ is constant, so $\Delta |F^+|^2 = 0$. Hence,

$$|\nabla F^+|^2 = 0 \quad \Rightarrow \quad \nabla F^+ = 0.$$

Thus, F^+ is parallel with respect to the Levi-Civita connection of g' .

Step 3: Modified scalar curvature is zero

Recall the modified scalar curvature:

$$\Phi_{\hat{g}} = R_{\hat{g}} - 2\sqrt{6} |W^+|_{\hat{g}} - 3\gamma_1 |F^+|_{\hat{g}}.$$

This quantity appeared in the definition of the modified Yamabe operator $L = -6\Delta + \Phi$. Equality in the energy inequality implies $\lambda_1(L) = 0$, and that L is minimized by a smooth, positive function (the Yamabe minimizer), so the scalar curvature in the conformal metric g' satisfies:

$$\Phi_{g'} \equiv 0.$$

Hence,

$$R_{g'} = 2\sqrt{6} |W^+|_{g'} + 3\gamma_1 |F^+|_{g'}.$$

This is a pointwise identity, not just an integral one, and reflects a perfect balance between geometric and gauge curvature in the equality case.

Step 4: Topological implications from the Bochner formula

To understand the final part of the theorem—that either $b_2^+(X^4) = 0$ when $\gamma_1 > 0$, or that all harmonic self-dual 2-forms are parallel when $\gamma_1 = 0$ —we analyze real-valued self-dual 2-forms under the Yamabe metric g' .

Let $\omega \in \Lambda_+^2(X, \mathbb{R})$ be a harmonic self-dual 2-form. Then the classical Bochner-Weitzenböck Formula (3.2.3) gives

$$\frac{1}{2}\Delta|\omega|^2 = |\nabla\omega|^2 - 2\langle W^+ \star \omega, \omega \rangle + \frac{1}{3}R|\omega|^2.$$

From the estimate for the Weyl tensor term (3.2.5), we obtain

$$\frac{1}{2}\Delta|\omega|^2 \geq |\nabla\omega|^2 + \frac{1}{3}(R - 2\sqrt{6}|W^+|)|\omega|^2.$$

But in the equality case, the modified scalar curvature vanishes pointwise, i.e.

$$R - 2\sqrt{6}|W^+| = 3\gamma_1|F^+| > 0 \quad \text{if } \gamma_1 > 0.$$

Hence

$$\frac{1}{2}\Delta|\omega|^2 \geq |\nabla\omega|^2 + \gamma_1|F^+||\omega|^2. \tag{3.31}$$

Now integrate over X . Since X is closed, the integral of $\Delta|\omega|^2$ vanishes, then (3.2.12) reads

$$0 \geq \int_X |\nabla\omega|^2 + \gamma_1|F^+||\omega|^2.$$

If $\gamma_1 > 0$ and $|F^+| > 0$, this implies that both $|\nabla\omega|$ and $|\omega|$ must vanish, so $\omega \equiv 0$, i.e., there are no nontrivial harmonic self-dual 2-forms. Hence $b_2^+(X) = 0$.

Now consider the special case when $\gamma_1 = 0$. Then the inequality (3.2.12) becomes

$$\frac{1}{2}\Delta|\omega|^2 \geq |\nabla\omega|^2,$$

which implies that $|\nabla\omega| \equiv 0$, so ω is parallel.

Illustrative Example: The BPST Instanton on $\mathbb{S}^4$

It is possible to achieve equality in (3.2.1) via the classic construction of BPST/ADHM $\mathrm{SU}(2)$ instantons on $\mathbb{S}^4$. The standard connection is expressed on $\mathbb{R}^4$, thought of as $\mathbb{S}^4 \setminus \{N\}$ as the $\mathfrak{su}(2)$ -valued 1-form

$$\theta = \frac{1}{1 + |x|^2}(\theta_1 \otimes \mathbf{i} + \theta_2 \otimes \mathbf{j} + \theta_3 \otimes \mathbf{k}),$$

where

$$\begin{aligned}\theta_1 &= x_1 dx^2 - x_2 dx^1 + x_3 dx^4 - x_4 dx^3 \\ \theta_2 &= x_1 dx^3 - x_3 dx^1 + x_4 dx^2 - x_2 dx^4 \\ \theta_3 &= x_1 dx^4 - x_4 dx^1 + x_2 dx^3 - x_3 dx^2.\end{aligned}$$

and

$$\mathbf{i} = \left(\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right), \quad \mathbf{j} = \left(\begin{array}{cc|c} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{array} \right), \quad \mathbf{k} = \left(\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{array} \right).$$

A classic calculation yields the relevant curvature tensor

$$F_\nabla = \frac{2}{(1+|x|^2)^2} \{ (dx^{12} + dx^{34}) \otimes \mathbf{i} + (dx^{13} - dx^{24}) \otimes \mathbf{j} + (dx^{14} + dx^{23}) \otimes \mathbf{k} \}.$$

Using our conventions for the metric induced by the Euclidean metric on two-forms, $|dx^{12}|^2 = 2$, etc. Moreover, using the inner product on $\mathfrak{g}_E$, it follows that $|\mathbf{i}|^2 = |\mathbf{j}|^2 = |\mathbf{k}|^2 = 2$. Putting these together we can compute the pointwise norm of F_∇ in this context to yield

$$|F_\nabla|^2 = \frac{4}{(1+|x|^2)^4} \left(2|dx^{12} + dx^{34}|^2 + 2|dx^{13} - dx^{24}|^2 + 2|dx^{14} + dx^{23}|^2 \right) = \frac{96}{(1+|x|^2)^4}.$$

Using the conformal invariance of the Yang–Mills energy, we thus obtain

$$\|F_\nabla\|_{L^2(\mathbb{S}^4, g_{\mathbb{S}^4})}^2 = 96 \int_{\mathbb{R}^4} \frac{1}{(1+|x|^2)^4} dV_{Eucl} = 16\pi^2.$$

Having computed the Yang–Mills energy, we turn to the remaining quantities in (3.2.1). A well-know fact is that the Yamabe invariant of the round 4-sphere is $8\sqrt{6}\pi$. Thus, employing Lie-Algebraic Estimate (3.2.6) we observe that the right hand side of the right hand side of (3.2.1) can be estimated as

$$3\gamma_1 \|F_\nabla^+\|_{L^2} \leq 3 \left(\frac{2}{\sqrt{3}} \gamma_0 \right) (4\pi) \leq 8\sqrt{6}\pi = Y([g_{\mathbb{S}^4}]).$$

Thus (3.2.1) is an equality, as are all the intermediate estimates.

The above calculation verifies that the BPST instanton on $(\mathbb{S}^4, g_{\mathbb{S}^4})$ saturates the energy gap inequality stated in Theorem 3.2.1. As a result, all intermediate estimates used in the proof must also be equalities. This rigid equality chain reflects several remarkable geometric features of the BPST/ADHM construction:

- First, the curvature of this $\mathrm{SU}(2)$ instanton achieves equality in the key algebraic inequality involving the Lie bracket:

$$\langle \omega, [\omega, \omega] \rangle \leq \gamma_1 |\omega|^3,$$

realizing the optimal value $\gamma_1 = \frac{4\sqrt{3}}{3}$. This demonstrates the sharpness of the algebraic constant in the theorem.

- Second, the equality case forces *parallelism* of the self-dual curvature F_{∇}^+ with respect to a specific metric in the conformal class $[g]$. In this example, the distinguished BPST instanton connection has curvature that is already parallel with respect to the round metric g_{S^4} .
- Third, since all BPST/ADHM connections are related by conformal transformations (i.e., they lie in the same orbit under the conformal group action), it follows that any such instanton also has parallel curvature with respect to a metric in its conformal class. What the theorem reveals is that this metric arises naturally as the solution to a modified Yamabe problem—a variational method that not only proves the inequality but also constructs the distinguished conformal representative in a general setting.

In conclusion, this explicit example not only confirms the sharpness of the energy gap but also illustrates how geometric rigidity—parallelism, constant curvature norm, vanishing W^+ —emerges in the equality case. The BPST instanton serves as a canonical model demonstrating the deep link between Yang–Mills theory, conformal geometry, and scalar curvature optimization.

Applications and Corollaries

One immediate corollary is a universal energy bound for non-instanton Yang–Mills connections on conformally flat four-manifolds, that is

Corollary 3.11. *Let (X^4, g) be a closed, oriented, conformally flat four-manifold with $Y([g]) > 0$. Suppose ∇ is a Yang–Mills connection on a vector bundle E over X^4 with structure group $G \subset SO(E)$, and curvature F_{∇} . Then ∇ is either an instanton, or satisfies*

$$\int_X |F_{\nabla}|_g^2 dV_g \geq 4\pi^2 p_1(E) + \frac{2Y([g])^2}{9\gamma_1^2} \geq 4\pi^2 p_1(E) + \frac{Y([g])^2}{12}. \quad (3.32)$$

In particular,

1. *For $E \rightarrow (\mathbb{S}^4, g_{\mathbb{S}^4})$ an $SU(2)$ bundle, a Yang–Mills connection ∇ is either an instanton, or satisfies*

$$\int_{\mathbb{S}^4} |F_{\nabla}|_g^2 dV_g \geq 16\pi^2 |\kappa(E)| + 32\pi^2. \quad (3.33)$$

2. *For $E \rightarrow (\mathbb{S}^4, g_{\mathbb{S}^4})$ an $SO(3)$ bundle, a Yang–Mills connection ∇ is either an instanton, or satisfies*

$$\int_{\mathbb{S}^4} |F_{\nabla}|_g^2 dV_g \geq 16\pi^2 |\kappa(E)| + 64\pi^2. \quad (3.34)$$

Proof. We recall the fundamental Chern-Weil formula

$$4\pi^2 p_1(E) = \int_X (|F_{\nabla}^-|_g^2 - |F_{\nabla}^+|_g^2) dV_g. \quad (3.35)$$

Since we have assumed our metric is conformally flat and $Y([g]) > 0$, we may combine (3.2.1) and (3.2.16) to yield

$$\begin{aligned} \|F_{\nabla}\|_{L^2}^2 &= \|F_{\nabla}^-\|_{L^2}^2 + \|F_{\nabla}^+\|_{L^2}^2 \\ &= 4\pi^2 p_1(E) + 2 \int_X |F_{\nabla}^+|_g^2 dV_g \\ &\geq 4\pi^2 p_1(E) + \frac{2Y([g])^2}{9\gamma_1^2}, \end{aligned}$$

as claimed. For the special case of $(\mathbb{S}^4, g_{\mathbb{S}^4})$, as discussed before we know that the Yamabe invariant is $8\sqrt{6}\pi$. Furthermore, applying the previously discussed Lie-algebraic estimate, we know that in the case of structure group $\mathrm{SU}(2)$ we may choose $\gamma_1 = \frac{4}{\sqrt{6}}$, whereas for $\mathrm{SO}(3)$ one has $\gamma_1 = \frac{2}{\sqrt{3}}$. This yields the remaining statements. $\square$

Another powerful application concerns the long-time behavior of the Yang–Mills flow. Building on prior work on bubbling and energy concentration, the authors show that if a connection on the trivial bundle has energy below $16\pi^2$, the flow must exist globally and converge to a flat connection. Their result gives an explicit, sharp energy threshold for global existence—something not previously achievable without stronger geometric assumptions.

Broader Significance and Future Directions

This work represents a significant conceptual advance in Yang–Mills theory. It not only bridges curvature estimates with conformal geometry but also introduces a new paradigm for understanding non-self-dual Yang–Mills connections through global invariants like the Yamabe constant. The conformally invariant nature of the result allows it to transcend the limitations of prior local curvature positivity assumptions.

In terms of future work, natural directions include extending this framework to complete noncompact manifolds, exploring whether analogous results exist in dimensions beyond four, and further leveraging the modified Yamabe operator in related geometric variational problems. Moreover, the connection to Yang–Mills flow dynamics opens pathways for deeper study of singularity formation and stability criteria in gauge theory.

3 Preliminaries

In this section, we will give general discussions on Yang-Mills connections, particularly, the anti-self-dual instantons. And we will analyze the anti-self-duality in a few important cases. This section is sourced from Tian's paper, "*Gauge Theory and Calibrated Geometry, I*".

Lemma 3.12. *Let A be a unitary connection of E over M such that $\text{tr}(F_A)$ is a harmonic 2-form and*

$$\Omega \wedge (F_A - \frac{1}{r} \text{tr}(F_A) \text{Id}) = - * (F_A - \frac{1}{r} \text{tr}(F_A) \text{Id}), \quad (3.36)$$

then A is a Yang-Mills connection. Moreover, if M is a compact manifold without boundary,

$$\frac{1}{4\pi^2} \int_M |F_A|^2 dV_g - \frac{1}{4r\pi^2} \int_M |\text{tr}(F_A)|^2 dV_g = \left(2C_2(E) - \frac{r-1}{r} C_1(E)^2 \right) \cdot [\Omega], \quad (3.37)$$

where $[\Omega]$ denotes the cohomology class of Ω .

4 Stability and Isolation Phenomena for Yang-Mills Fields

Theorem 3.13. *Any weakly stable Yang-Mills field over S^4 with group $G = \text{SU}_2, \text{SU}_3$ or U_2 is either self-dual or anti-self-dual.*

Chapter 4 The Yang-Mills Flow

1 Basic Properties of Yang-Mills Flow

Local Existence, Uniqueness and ε -Regularity Theorem

2 Blow-up Analysis

3 Energy Identity

4 The Yang-Mills α -Flow

Chapter 5 Long-time existence and Smooth Convergence of Yang-Mills Flow

1 SD (ASD) Singularities

2 Sobolev and Poincaré Inequality for SD forms

3 Convergence at infinite time

4 Further Result

In this section we introduce Alex Waldron's work, "*Long-time existence for Yang-Mills flow*", which solved the existence of Yang-Mills flow. Main theorem established in this paper is:

Theorem 5.1. *Suppose that $A(t)$ is a smooth solution of (YM) over $M^4 \times [0, T)$, with $T < \infty$, and*

$$\sup_{0 \leq t < T} \int_M |F(t)|^2 dV + \int_0^T \int_M |D^* F|^2 dV dt < \infty. \quad (5.1)$$

For any $\epsilon > 0$ and $x \in M$, there exists $\lambda > 0$ such that

$$\limsup_{t \rightarrow T} \int_{B_\lambda(x)} |F(t)|^2 dV < \epsilon \quad (5.2)$$

Moreover, $\lim_{t \rightarrow T} A(t)$ exists in C_{loc}^∞ .

Corollary 5.2. *Assume M^4 is compact. Then any classical solution of (YM) extends smoothly for all time.*

For any Sobolev H^1 initial connection, there exists a weak solution defined on $M \times [0, \infty)$ which attains the initial data, in the sense of Struwe [17], and is smooth modulo gauge for $t > 0$.¹

Chapter 6 Hermitian Yang-Mills Flow

1 Basic Properties of Hermitian Yang-Mills Flow